



PROJECT REPORT

Customer Shopping Trends Analysis



Prepared by:

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SIVADASAN**

For:

CIS4008-N-CF1-2023

**BIG DATA AND BUSINESS
INTELLIGENCE**

January, 2024

Teesside University, United Kingdom

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SECTION 1

BUSINESS INTELLIGENCE DESIGN

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A. BI Data Source Description and Questions

Data Source Description:

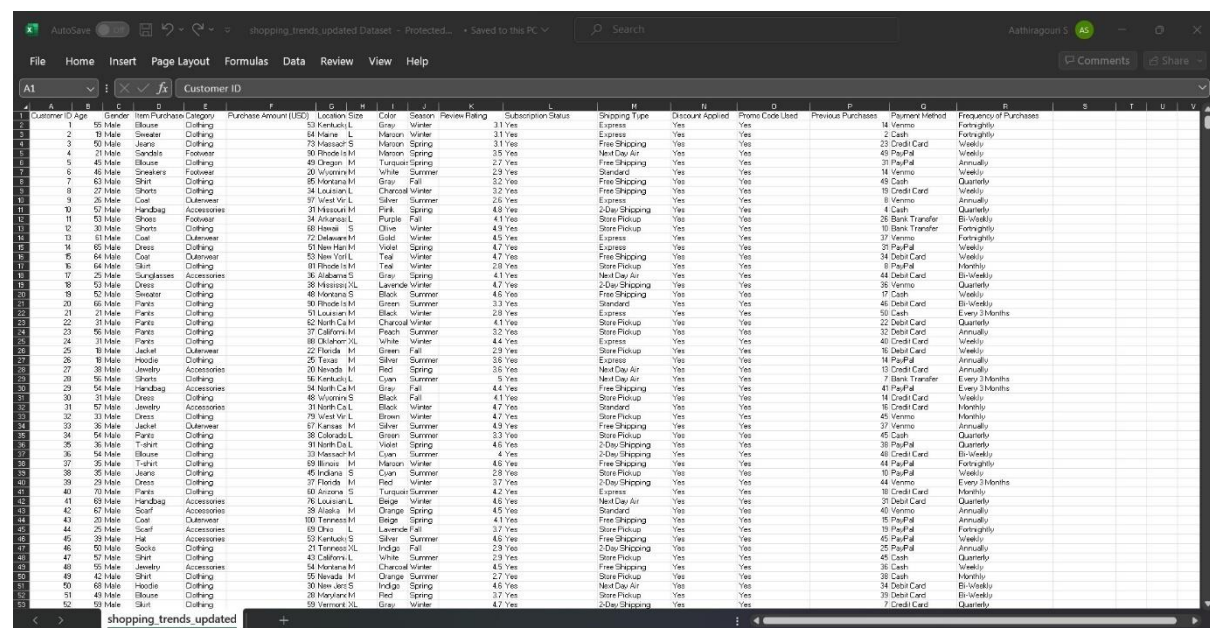
Corporate and business sector, has forever been one of the most rapidly transforming sectors ever. As new ventures are being launched on a regular basis, retail, e-Commerce and associated domains continue to be relevant and is among the leading industries in the present business scene worldwide. Due to the elevated levels of competition in this industry, Data Analytics and Business Intelligence solutions, developed using advanced technology is the key to succeed.

The **Customer Shopping Trends Dataset**, initially named “*shopping_trends_updated*”, illustrates the data of a typical Apparel Shopping Website, that sells its products through physical stores as well as through customer shopping websites. This dataset is an ideal choice to study general consumer behaviour in apparel retail industry and analysing this dataset can provide deep and meaningful insights into the operation and overall characteristics of the retail industry.

The dataset has been obtained from the following source:

<https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset/data>

The following shows the dataset in its original excel format.



Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
1	35	Male	Blouse	Clothing	53	Kanku	L	Grey	Winter	3.1	Yes	Express	Yes	Yes	14	Vermo	Fortnightly
2	19	Male	Sweater	Accessories	44	Harve	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
3	50	Male	Jeans	Clothing	79	Kapasa	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
4	21	Male	Sandals	Footwear	90	Rhodie	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
5	45	Male	Blouse	Clothing	45	Oragan	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually
6	45	Male	Sneakers	Footwear	30	Vijayana	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Vermo	Weekly
7	62	Male	Shirt	Clothing	95	Kanana	M	Grey	Fall	3.2	Yes	Free Shipping	Yes	Yes	40	Cash	Quarterly
8	27	Male	Shorts	Clothing	34	Louisa	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	Credit Card	Weekly
9	26	Male	Coat	Outerwear	37	Vest Vi	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Vermo	Annually
10	57	Male	Handbag	Accessories	71	Hassou	M	Pink	Spring	4.9	Yes	2-Day Shipping	Yes	Yes	1	Cash	Quarterly
11	53	Male	Shoes	Footwear	34	Akrona	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	Yes	26	Bank Transfer	Bi-Weekly
12	38	Male	Shorts	Clothing	69	Hewai	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	Yes	19	Bank Transfer	Fortnightly
13	51	Male	Coat	Outerwear	72	Deawen	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	37	Vermo	Fortnightly
14	65	Male	Coat	Outerwear	91	New Har	M	Violet	Winter	4.7	Yes	Express	Yes	Yes	97	PayPal	Weekly
15	65	Male	Coat	Outerwear	53	New Yori	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	Yes	34	Debit Card	Weekly
16	64	Male	Skirt	Clothing	91	Rhodie	M	Teal	Winter	2.8	Yes	Next Day Air	Yes	Yes	38	PayPal	Monthly
17	25	Male	Sunglasses	Accessories	36	Alakana	S	Grey	Spring	4.1	Yes	Next Day Air	Yes	Yes	41	Debit Card	Bi-Weekly
18	53	Male	Dress	Clothing	38	Mipassa	L	Lavender	Winter	4.7	Yes	2-Day Shipping	Yes	Yes	36	Vermo	Quarterly
19	62	Male	Sweater	Clothing	46	Vijana	S	Black	Summer	4.6	Yes	Free Shipping	Yes	Yes	17	Cash	Weekly
20	66	Male	Pants	Clothing	90	Rhodie	M	Green	Summer	3.3	Yes	Standard	Yes	Yes	46	Debit Card	Bi-Weekly
21	21	Male	Pants	Clothing	51	Louisa	M	Black	Winter	2.9	Yes	Express	Yes	Yes	50	Cash	Every 3 Months
22	21	Male	Pants	Clothing	52	North Cal	M	Charcoal	Winter	4.1	Yes	Store Pickup	Yes	Yes	22	Debit Card	Quarterly
23	55	Male	Pants	Clothing	37	California	M	Peach	Summer	3.2	Yes	Store Pickup	Yes	Yes	32	Debit Card	Annually
24	31	Male	Pants	Clothing	80	Decker	L	White	Winter	4.4	Yes	Express	Yes	Yes	40	Credit Card	Weekly
25	18	Male	Jacket	Outerwear	22	Florida	M	Green	Fall	2.9	Yes	Store Pickup	Yes	Yes	16	Debit Card	Weekly
26	18	Male	Hoodie	Clothing	25	Texas	M	Silver	Summer	3.6	Yes	Express	Yes	Yes	14	PayPal	Annually
27	38	Male	Jewelry	Accessories	30	Nevada	M	Red	Spring	3.6	Yes	Next Day Air	Yes	Yes	13	Credit Card	Annually
28	56	Male	Shorts	Clothing	56	Kanku	L	Cyan	Summer	5	Yes	Next Day Air	Yes	Yes	7	Bank Transfer	Every 3 Months
29	54	Male	Handbag	Accessories	54	North Cal	M	Grey	Fall	6.4	Yes	Free Shipping	Yes	Yes	49	PayPal	Every 3 Months
30	31	Male	Dress	Clothing	46	Vijana	S	Black	Fall	4.1	Yes	Store Pickup	Yes	Yes	14	Credit Card	Weekly
31	57	Male	Jewelry	Accessories	71	North Cal	L	Black	Winter	4.7	Yes	Standard	Yes	Yes	18	Credit Card	Monthly
32	33	Male	Dress	Clothing	79	Vest Vi	L	Brown	Winter	4.7	Yes	Store Pickup	Yes	Yes	45	Vermo	Monthly
33	36	Male	Jacket	Outerwear	57	Kanku	S	Silver	Summer	4.9	Yes	Free Shipping	Yes	Yes	37	Vermo	Annually
34	54	Male	Pants	Clothing	36	Calakadi	L	Green	Summer	5.3	Yes	2-Day Shipping	Yes	Yes	45	Cash	Quarterly
35	36	Male	T-shirt	Clothing	91	North Cal	L	Violet	Spring	4.6	Yes	2-Day Shipping	Yes	Yes	38	PayPal	Quarterly
36	54	Male	Blouse	Clothing	34	Harve	M	Cyan	Summer	4	Yes	2-Day Shipping	Yes	Yes	41	Credit Card	Bi-Weekly
37	25	Male	T-shirt	Clothing	69	Illinois	M	Maroon	Winter	4.6	Yes	Free Shipping	Yes	Yes	44	PayPal	Fortnightly
38	25	Male	Jeans	Clothing	45	Indiana	S	Cyan	Summer	2.8	Yes	Free Shipping	Yes	Yes	10	PayPal	Weekly
39	29	Male	Shoes	Footwear	37	Florida	M	Red	Winter	3.7	Yes	2-Day Shipping	Yes	Yes	44	Vermo	Every 3 Months
40	73	Male	Pants	Clothing	40	Arizona	S	Turquoise	Summer	4.2	Yes	Express	Yes	Yes	19	Credit Card	Monthly
41	69	Male	Handbag	Accessories	76	Louisa	L	Beige	Winter	4.6	Yes	Next Day Air	Yes	Yes	30	Debit Card	Quarterly
42	67	Male	Scarf	Accessories	38	Alaska	M	Orange	Spring	4.5	Yes	Standard	Yes	Yes	40	Vermo	Annually
43	25	Male	Coat	Outerwear	80	Tennessee	M	Beige	Fall	4.1	Yes	Free Shipping	Yes	Yes	15	PayPal	Yearly
44	25	Male	Hat	Accessories	69	Orio	L	Leopard	Spring	3.7	Yes	Free Shipping	Yes	Yes	16	PayPal	Fortnightly
45	39	Male	Hat	Accessories	53	Kanku	S	Silver	Summer	4.6	Yes	Free Shipping	Yes	Yes	45	PayPal	Weekly
46	60	Male	Socks	Clothing	21	Tennessee	L	Indigo	Fall	2.9	Yes	2-Day Shipping	Yes	Yes	25	PayPal	Annually
47	57	Male	Shirt	Clothing	43	California	L	White	Summer	2.9	Yes	Store Pickup	Yes	Yes	45	Cash	Quarterly
48	51	Male	Jewelry	Accessories	54	Kanana	M	Charcoal	Winter	4.5	Yes	Free Shipping	Yes	Yes	35	Cash	Weekly
49	42	Male	Shirt	Clothing	55	Nevada	M	Orange	Summer	2.7	Yes	Store Pickup	Yes	Yes	38	Cash	Monthly
50	68	Male	Hoodie	Clothing	30	New Jersey	S	Indigo	Spring	4.6	Yes	Next Day Air	Yes	Yes	34	Debit Card	Bi-Weekly
51	48	Male	Blouse	Clothing	39	Nevada	M	Red	Spring	3.7	Yes	Store Pickup	Yes	Yes	19	Debit Card	Bi-Weekly
52	59	Male	Skirt	Clothing	69	Vermont	L	Grey	Winter	4.7	Yes	2-Day Shipping	Yes	Yes	7	Credit Card	Quarterly

Figure 1: Customer Shopping Trends Dataset Excel file.

The factors featured in this dataset are defined as follows:

Table 1: Dataset Description

Column Name	Description
Customer ID	It is the unique identification number allocated to each customer. This column is of integer datatype.
Age	This column describes the age, in years, of each customer. It is of integer datatype.
Gender	This column describes gender of the customer as either 'male' or 'female'. It is a non-numeric column.
Item Purchased	This column describes type of apparel item purchased by each customer, such as 'Gloves', 'Sweater', 'Boots' etc.
Category	This column is used to classify each value in the 'Item Purchased' column into the apparel category it belongs to such as 'Accessories', 'Outerwear' etc.
Purchase Amount (USD)	The amount received for each sold item, in USD, is described by this column.
Location	The residing city/state in the USA for each customer is described by this column.
Size	This column describes the size range in which each sold item falls in. The sizes available for all items are S, M, L and XL.
Colour	The colour classification is for each item sold is defined by this column.
Season	The season in which each item was sold is described by this column. The major classes are: Summer, Winter, Spring and Fall
Review Rating	Rating given by each customer after an item was sold to them is defined by this column. The rating is out of 5.00 and can take fractional values.
Subscription Status	This column describes the subscription status of each customer. The values are non-numerical and boolean (Yes/No).
Shipping Type	The shipping mode used to deliver the purchased item for each customer is described in this column.
Discount Applied	This column describes whether the item was sold to the concerned customer after applying any discount and so the 'Purchase

	Amount' column values are final amount received with or without any discount on the concerned item. The values are non-numerical and boolean (Yes/No).
Promo Code Used	This column describes whether the item was sold to each customer after a promo code was used. The values are non-numerical and boolean (Yes/No).
Previous Purchases	This column describes the number of times each customer has shopped on this website or shopped from this company, before the current purchase. It is a numerical column.
Payment Method	This column describes which payment mode was used for purchase by each customer.
Frequency of Purchases	This column classifies the general frequency with which each customer shops into classes such as: 'Weekly', 'Monthly', 'Annually' etc.

BI Scope and Objectives:

The BI scope and objectives for this project can be defined as:

- To observe customer purchase patterns and seasonal sales trends.
- To gain insights into the popularity of the business amongst customers and review overall ratings for understanding customer satisfaction levels.
- Identifying and monitoring the main KPIs (Key Performance Indicators) for this business.
- Analysing trending Categories and Items.
- Improving data visualisation and business decision-making.
- Developing a user-friendly dashboard for analysing the huge volume of business data.

BI Questions:

For visualising and analysing this data, some important Business Questions were framed and the answers to these can reveal furthermore details and insights into the data. The BI tools of Power BI can also be used to design a powerful report/dashboard that can aid in answering and elaborating on these topics in the best possible manner.

Since, the dataset is focused on sales, related KPIs and aspects needs to be examined.

The Business Intelligence Questions are as follows:

- What is the Total Sales Observed for customers who has a frequency of shopping Weekly?
- What is the Total Number of Items sold to customers who has a frequency of shopping Weekly?
- What is the Number of Subscriptions observed for customers who has a frequency of shopping Weekly?
- Which is the most sold Item to customers who has a frequency of shopping Annually?
- How many items were sold in each Size range?
- Which Gender type has recorded the greatest number of subscriptions?
- What is the Total Sales observed, Total Number of Items sold and Average Customer Review Rating?
- What is the Total Sales recorded for customers with and without Subscription?
- Which Season saw the greatest amount of Total Sales?

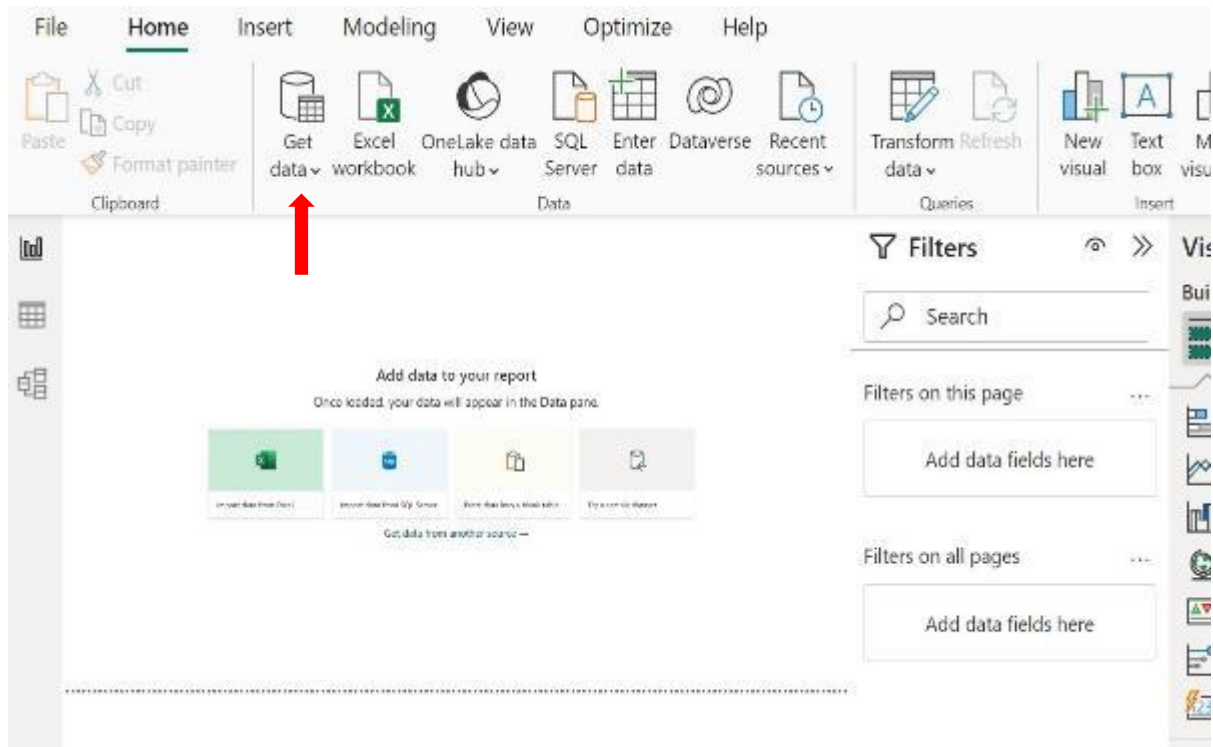
Rationale:

The chosen dataset is ideal for analysing retail consumer behaviour and sales patterns. This dataset is a window to understanding business processes involved in the industry. The general focus of this case would definitely be on increasing sales and improving selling patterns. The number of Items and Categories sold is a very important aspect of businesses like retail and eCommerce as they are competitive industries, where the quantity of units sold matter very much. Such aspects form the KPIs that need to be analysed. Another point to consider would be “understanding the customer base and their behaviours”. The factors such as review ratings, number of previous purchases and number of subscriptions indicate customer loyalty and customer satisfaction and analysing these would help in effective decision-making.

B. Data Preprocessing and Cleansing

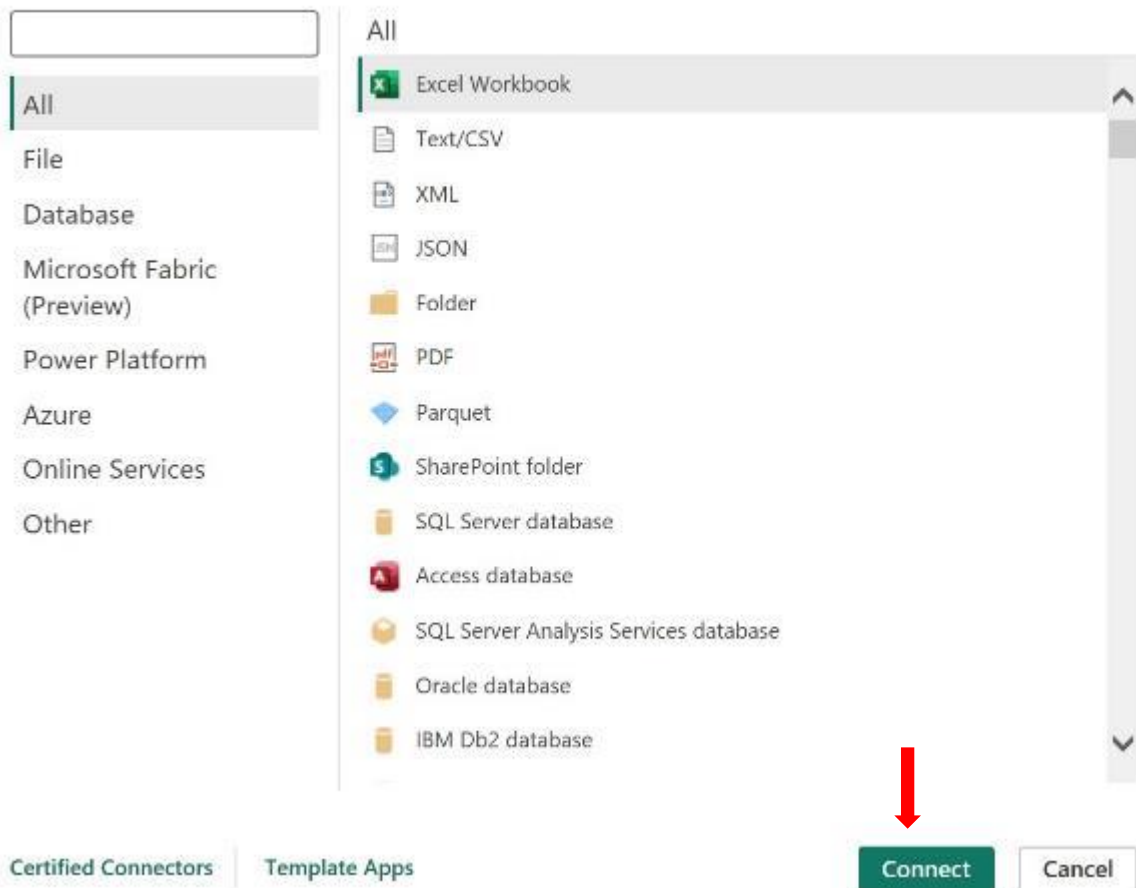
Loading Data:

The first step was to load the data into PowerBI. Hence the Get Data button was selected:



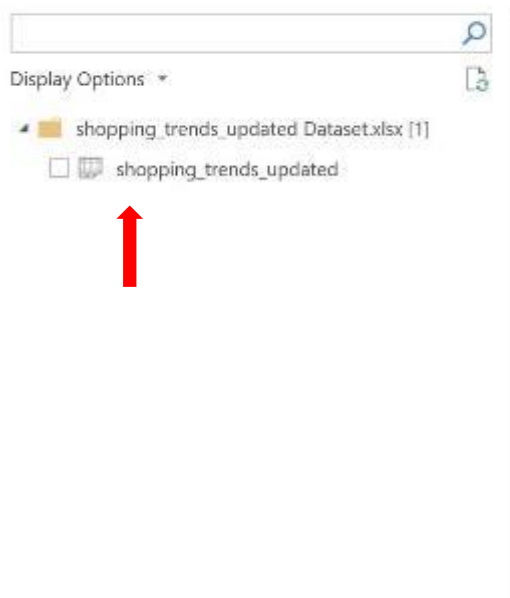
The following tab appears and here since the file to be loaded was an excel file, the 'Excel Workbook' option was selected.

Get Data



As soon as the 'Connect' button was selected, the following Navigator tab appeared. Indicated below are the visuals that show how the '*shopping_trends_updated*' file was selected and loaded into the Navigator.

Navigator



No items selected for preview

Navigator

shopping_trends_updated

Display Options ▾

- shopping_trends_updated Dataset.xlsx [1]
- ☒ shopping_trends_updated

Customer ID	Age	Gender	Item Purchased	Category	Purchase Price
1	55	Male	Blouse	Clothing	
2	29	Male	Sweatshirt	Clothing	
3	50	Male	Jeans	Clothing	
4	23	Male	Sandals	Footwear	
5	45	Male	Blouse	Clothing	
6	46	Male	Sneakers	Footwear	
7	68	Male	Shirt	Clothing	
8	27	Male	Shorts	Clothing	
9	26	Male	Coat	Outerwear	
10	57	Male	Handbag	Accessories	
11	53	Male	Shoes	Footwear	
12	30	Male	Shorts	Clothing	
13	62	Male	Coat	Outerwear	
14	65	Male	Dress	Clothing	
15	64	Male	Coat	Outerwear	
16	64	Male	Skirt	Clothing	
17	25	Male	Sunglasses	Accessories	

Load Transform Data Cancel

The image below shows the columns of the file loaded into PowerBI.

File Home Insert Modeling View Optimize Help

Clipboard: Cut, Copy, Paste, Format painter

Data: Get data, Excel workbook, OneLake data hub, SQL Server data, Enter data, Datasource, Recent sources

Queries: Transform Refresh data

Visuals: New visual, Text box, More visuals, New measure, Quick measure, Sensitivity, Publish

Build visuals with your data

Select or drag fields from the Data pane onto the report canvas.

Visualizations

Build visual

Filters

Values

Add data fields here

Drill through

Data

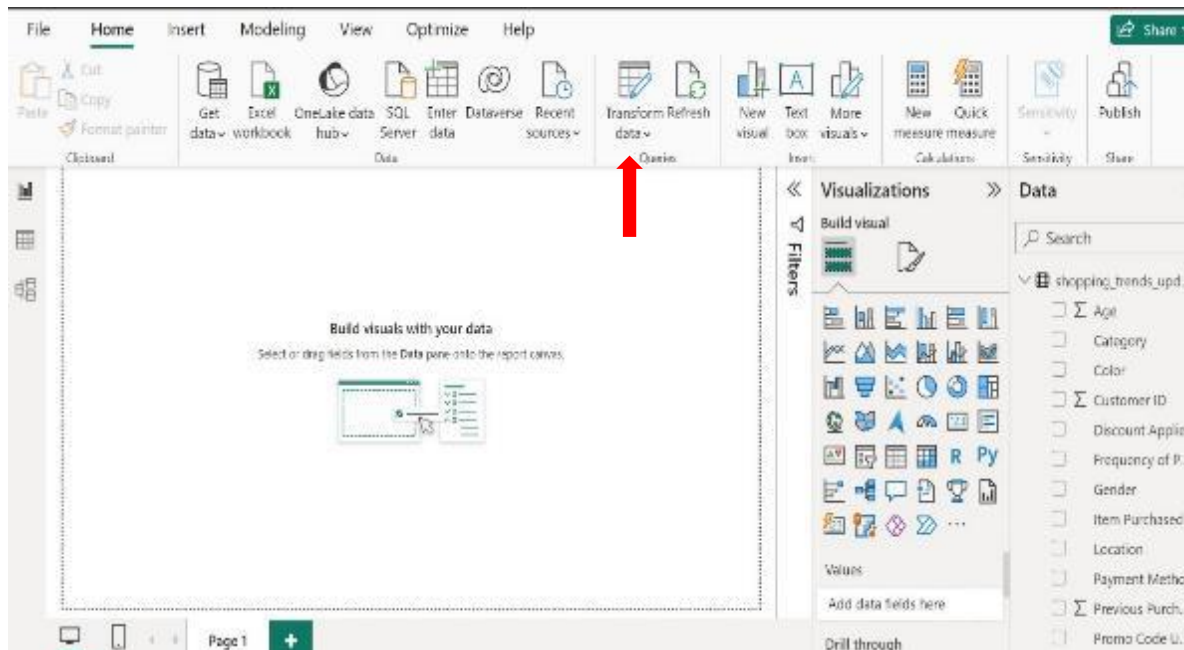
Search

shopping_trends_upd.

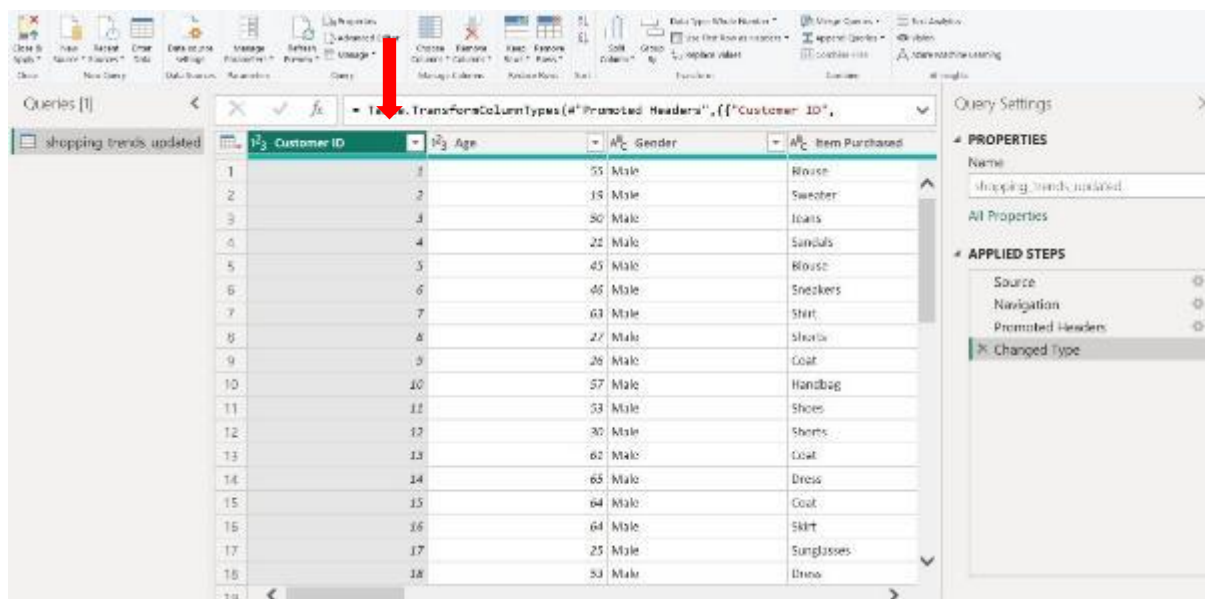
- ☐ Age
- ☐ Category
- ☐ Color
- ☐ Customer ID
- ☐ Discount Applied
- ☐ Frequency of Purchase
- ☐ Gender
- ☐ Item Purchased
- ☐ Location
- ☐ Payment Method
- ☐ Previous Purchase
- ☐ Promo Code Used

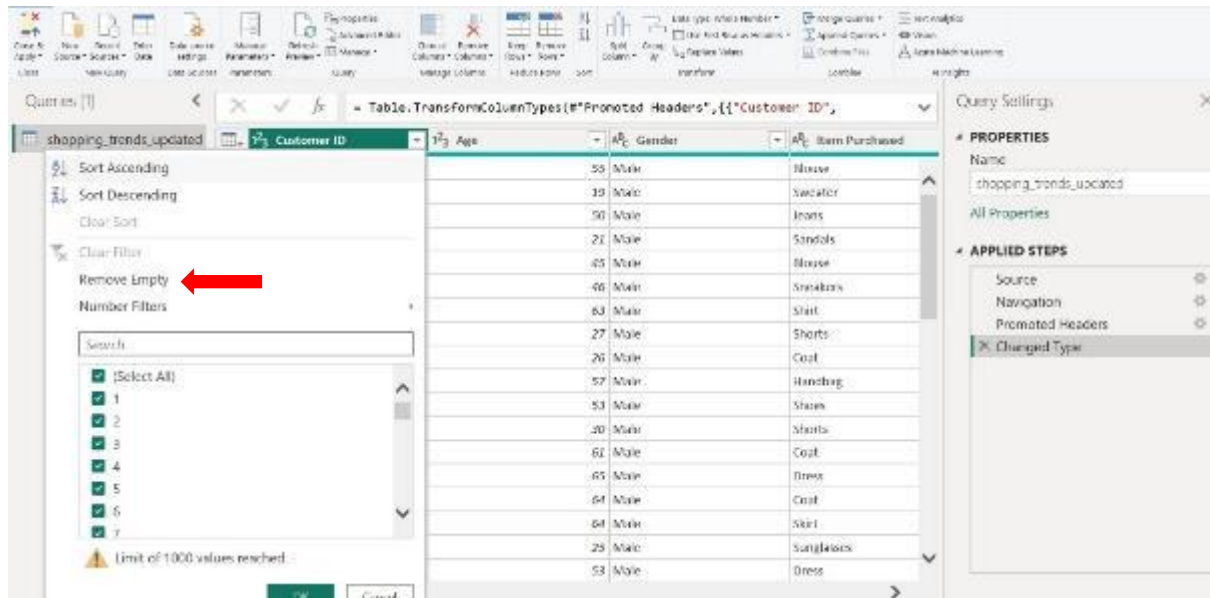
Data Cleansing and Preparation:

The Data Cleansing and Preparation step was started off by selecting the 'Transform Data' button as shown below.

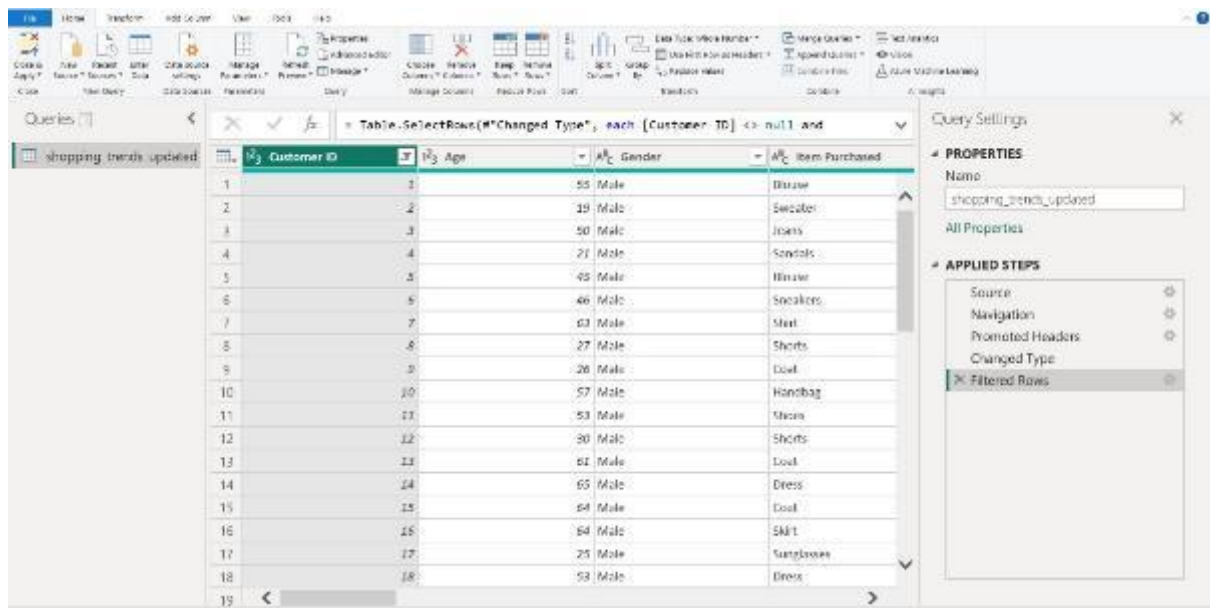


The following window, known as the 'Power Query Editor' appears where the following steps were carried out on the first column: 'Customer ID':





Initially, the dropdown was visually inspected for any 'null' or 'NA' values, but none were found. Hence, the 'Remove Empty' option was selected to omit any empty entries. The window looks as follows:



Next was to remove redundancy and errors. Hence, the 'Remove Duplicates' and 'Remove Errors' options were selected from the drop-down box that appeared after right-clicking on the column header:

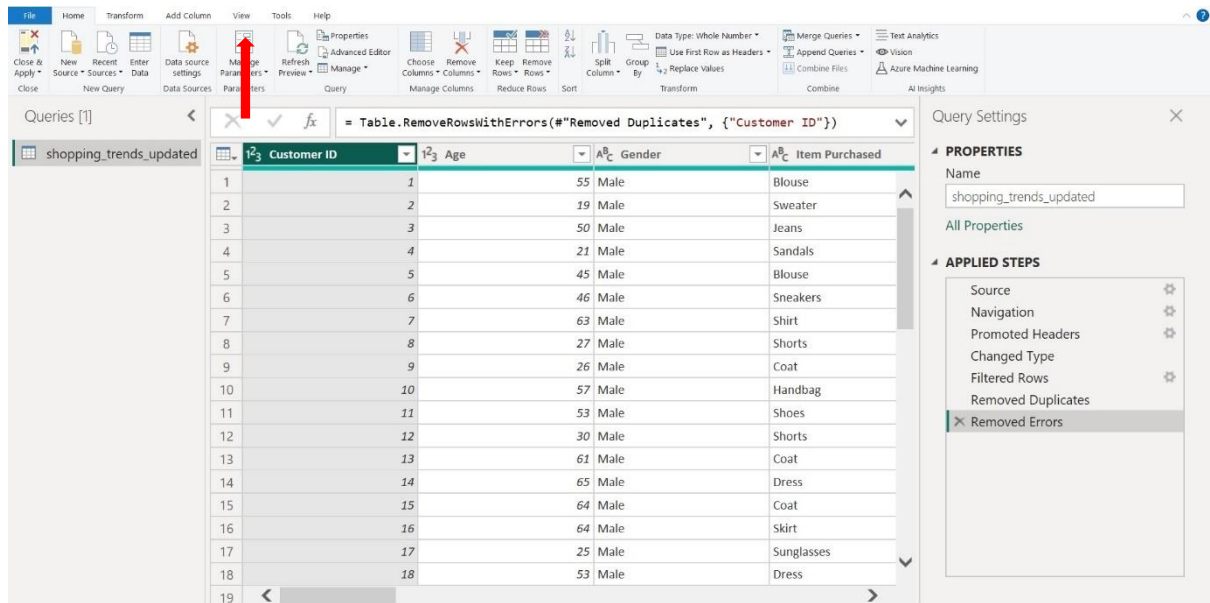
Table: RemoveRowsWithErrors(#"Removed Duplicates", ("Customer ID"))

Customer ID	Age	Gender	Item Purchased
1	2	55 Male	Blouse
2	3	19 Male	Sandwich
3	3	50 Male	Jeans
4	4	21 Male	Sandwich
5	5	40 Male	Blouse
6	5	40 Male	Sneakers
7	7	63 Male	Shirt
8	8	27 Male	Shirts
9	9	26 Male	Coat
10	10	57 Male	Handbag
11	12	53 Male	Shoes
12	12	30 Male	Shirts
13	13	61 Male	Coat
14	14	61 Male	Dress
15	15	64 Male	Coat
16	16	66 Male	Skirt
17	17	25 Male	Sunglasses
18	18	53 Male	Dress

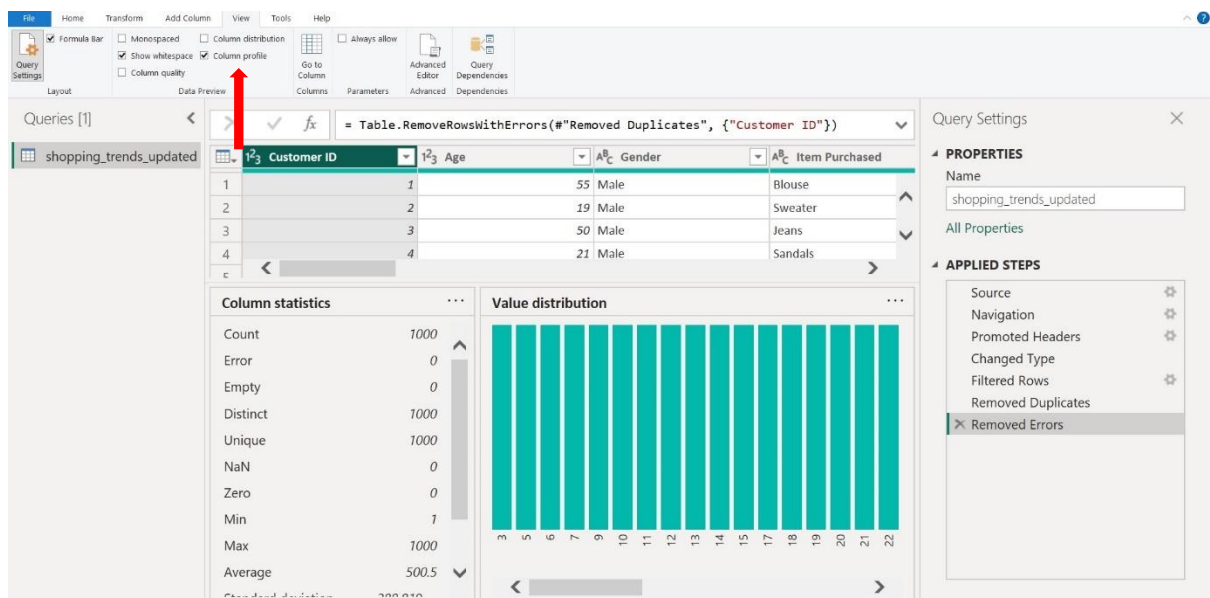
Table: RemoveRowsWithErrors(#"Removed Duplicates", ("Customer ID"))

Customer ID	Age	Gender	Item Purchased
1	2	55 Male	Blouse
2	3	19 Male	Sandwich
3	3	50 Male	Jeans
4	4	21 Male	Sandwich
5	5	40 Male	Blouse
6	5	40 Male	Sneakers
7	7	63 Male	Shirt
8	8	27 Male	Shirts
9	9	26 Male	Coat
10	10	57 Male	Handbag
11	12	53 Male	Shoes
12	12	30 Male	Shirts
13	13	61 Male	Coat
14	14	61 Male	Dress
15	15	64 Male	Coat
16	16	66 Male	Skirt
17	17	25 Male	Sunglasses
18	18	53 Male	Dress

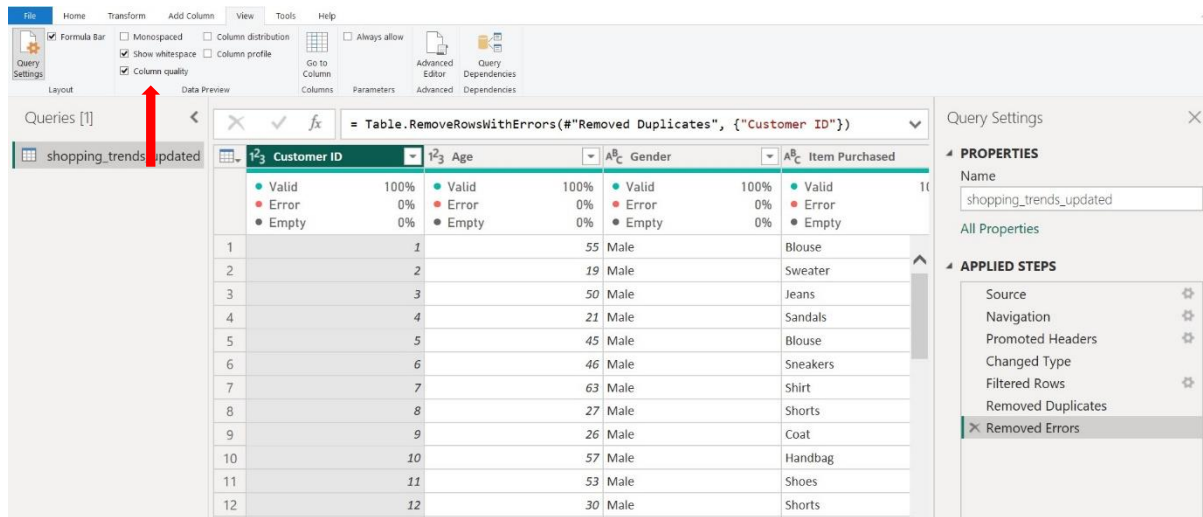
To further analyse the column, the 'View' tab was selected as shown below:



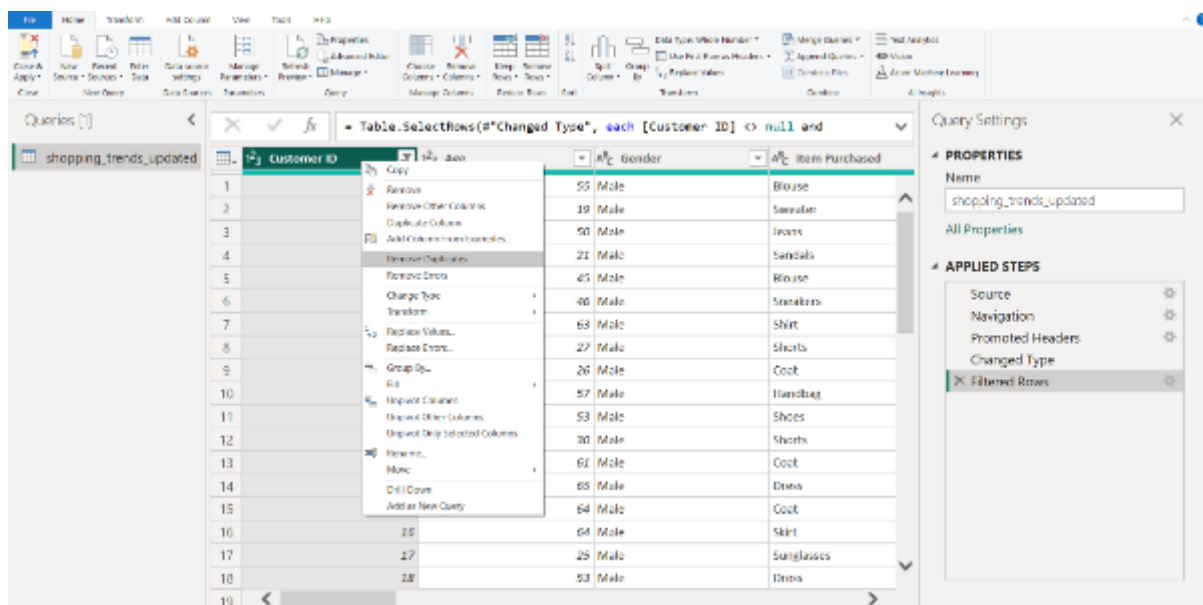
In this tab, the 'Column Profile' button was selected to see details as shown:



Similarly, after deselecting the 'Column Profile' button, the 'Column Quality' button was selected to reveal the following information:

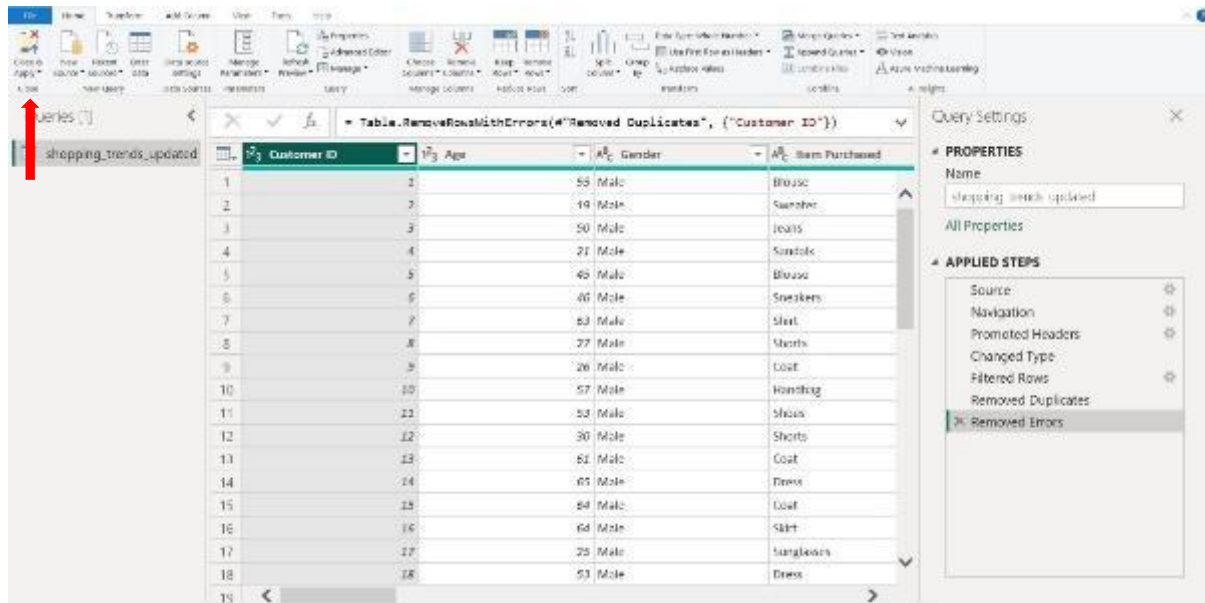


Next the datatypes of each column were inspected. Since, all the types were accurate, the types were not changed. However, the method to change type is illustrated as follows:



These steps were performed for all the columns to finally arrive at the cleansed and pre-processed dataset ready for visualisation.

To return to 'Report View', the 'Close and Apply' button was selected so that all the changes made in the 'Power Query Editor' were saved:



C. Data Modelling: Star Schema Facts and Dimensions

Generating Fact and Dimension Tables:

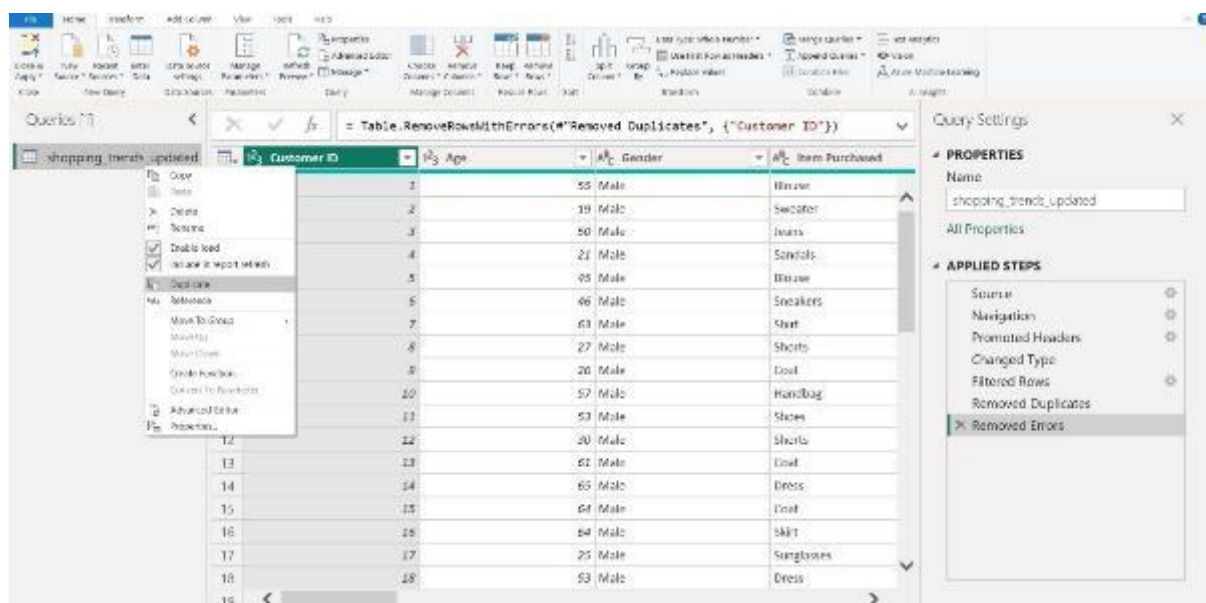
Aiming to reduce data redundancy and improve storage alongside better efficiency, the main table: 'shopping_trends_updated' table has been split into smaller tables such that there is a main table or 'Fact Table' and several smaller 'Dimensions Tables'.

a. Creating DiscountStatus Table

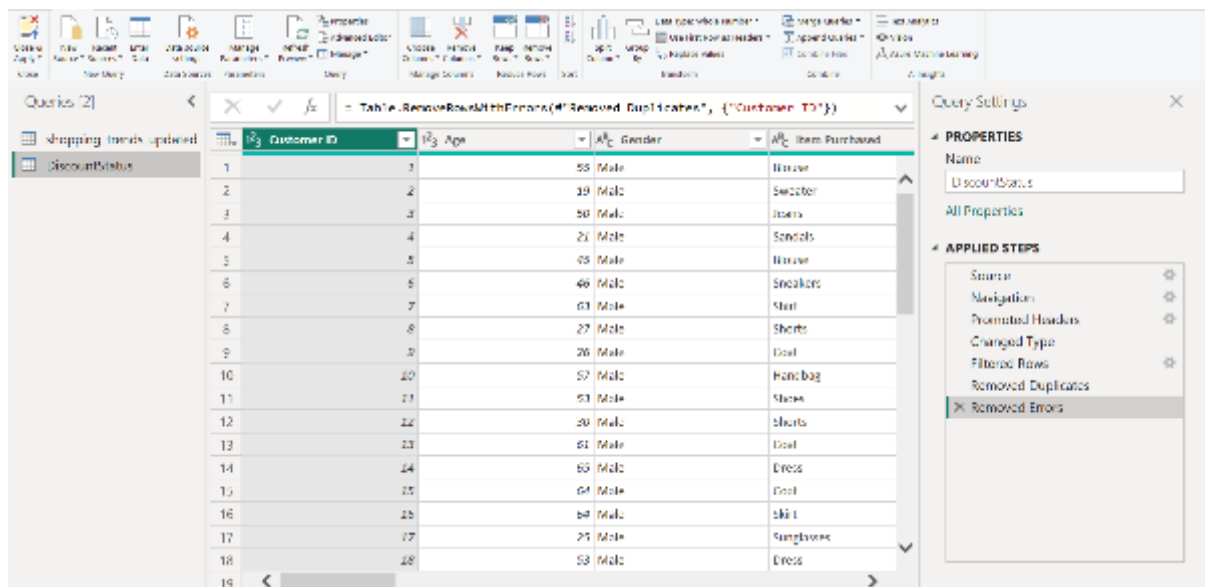
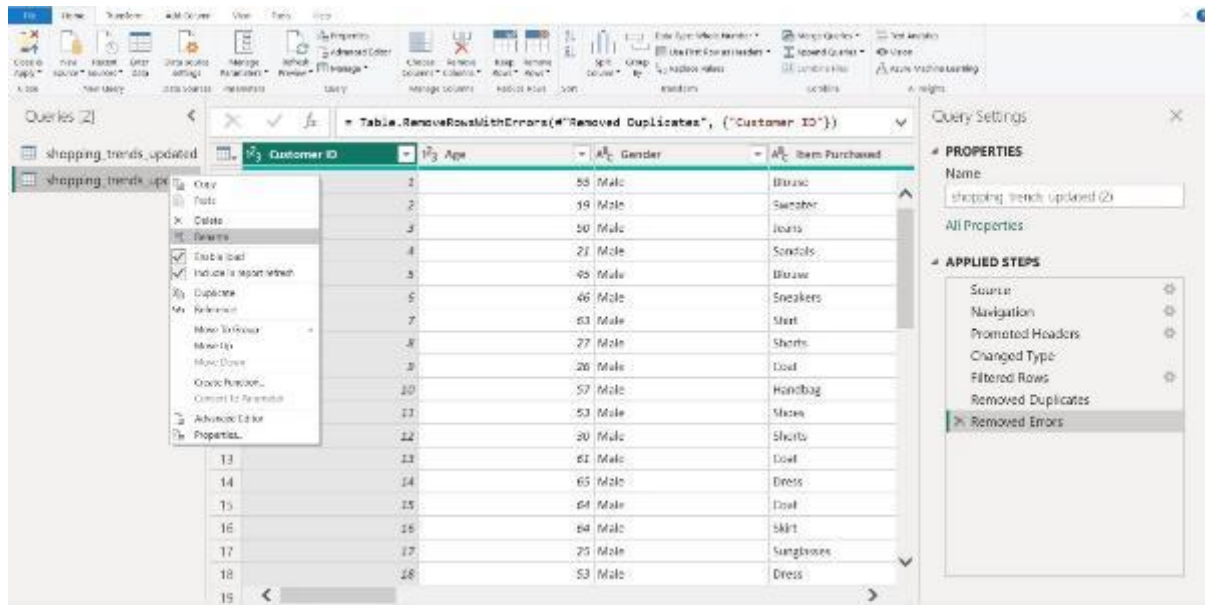
The 'DiscountStatus' table is used to access the information on whether the sale was made after a discount or not. This information is directly related to 'Promo Code Used' column, hence, the table shall be referenced using the 'Promo Code Used' as it would form the Primary Key in the 'DiscountStatus' table.

For creating a new table, the 'Transform Data' button was selected and the 'Power Query Editor' appeared.

Then, a duplicate of the 'shopping_trends_updated' table was created, by right-clicking on the table name as shown:

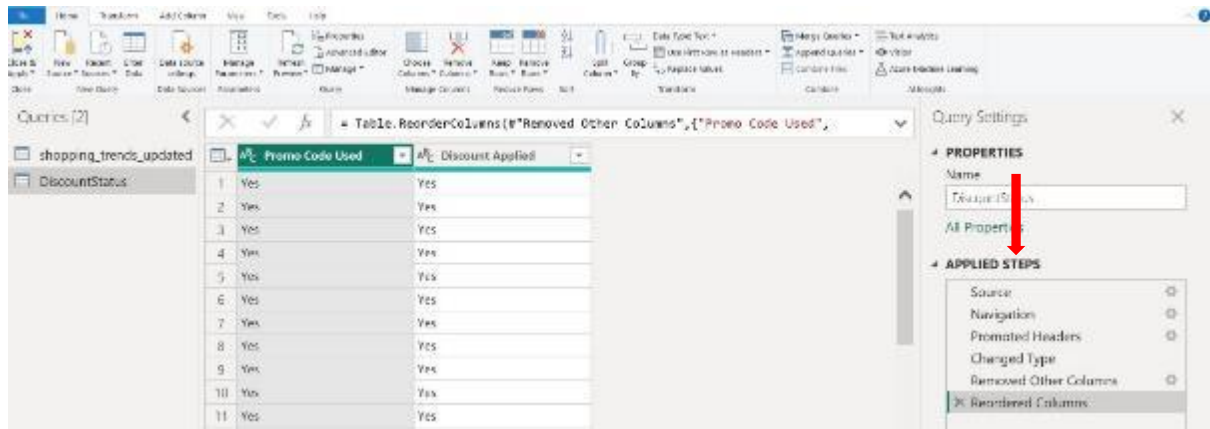


Then, this newly created duplicate 'shopping_trends_updated' was renamed as 'DiscountStatus' table, as shown below:

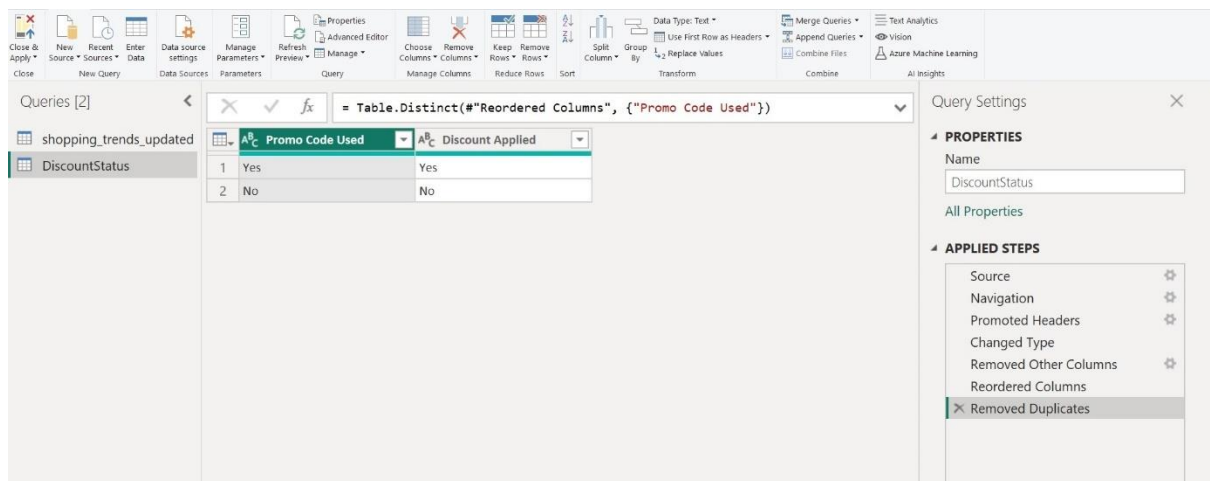


This is the very important and primary step performed to create all of the upcoming dimensions tables.

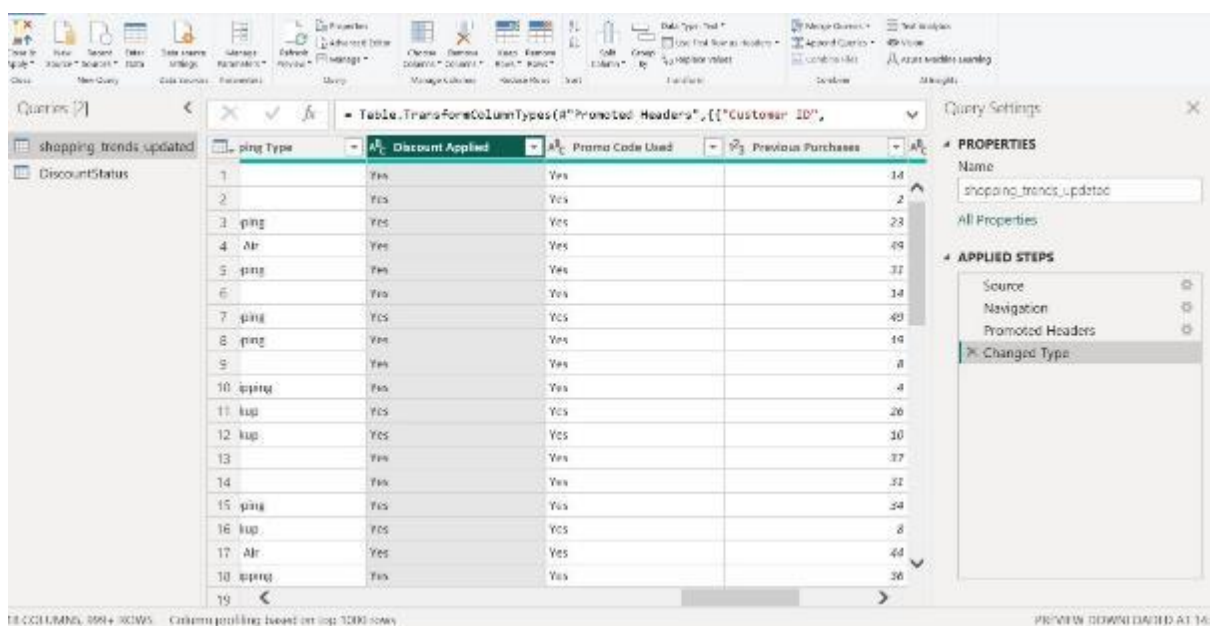
The next step was to remove all columns from 'DiscountStatus' table except for 'Discount Applied' column and 'Promo Code Used' which were then reordered as shown in the image below. The 'Applied Steps' window show that the aforementioned steps were performed.



Next, the 'Remove Duplicates' option was performed to obtain the following table:



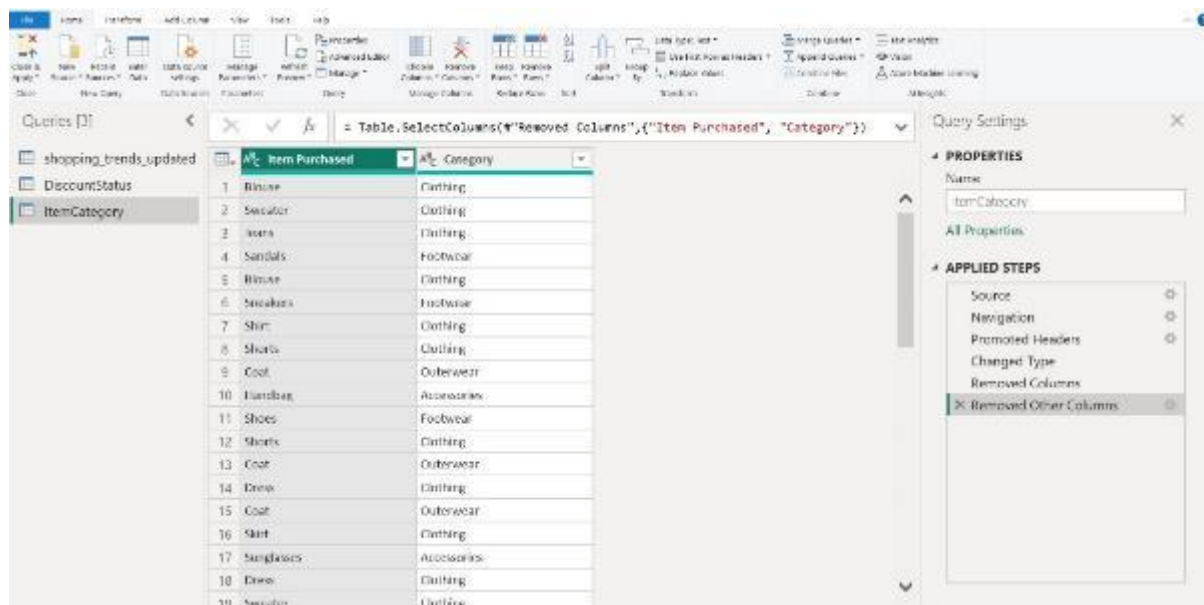
Finally, the 'Discount Applied' column was removed from the 'shopping_trends_updated' table, while retaining the 'Promo Code Used' column as it is the Foreign Key.



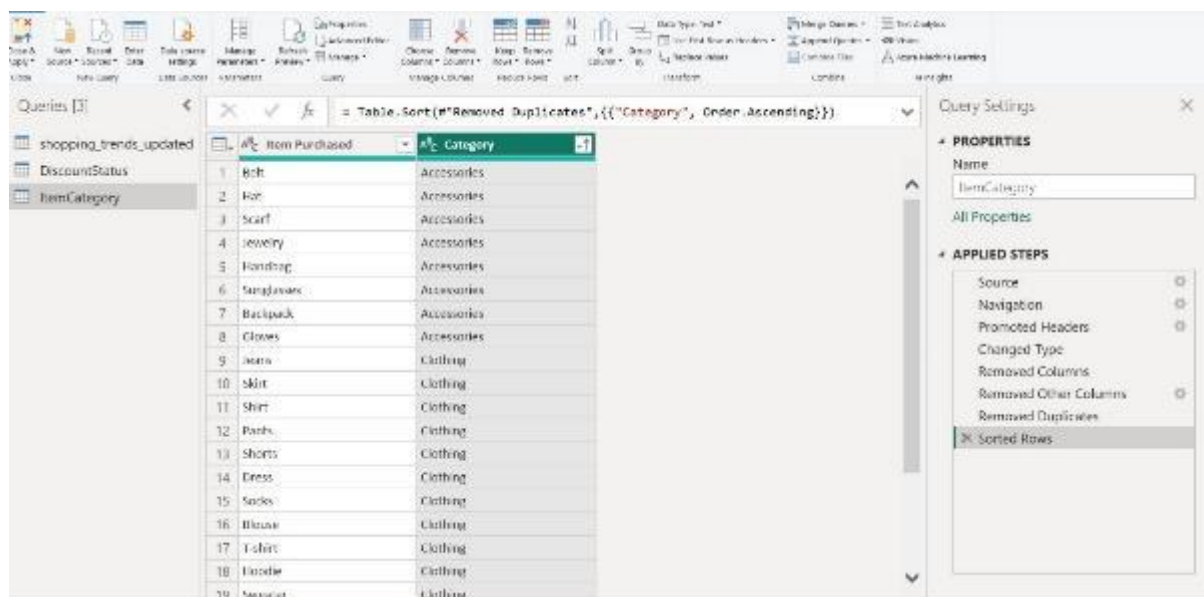
b. Creating ItemCategory Table

The 'ItemCategory' table is used to access the information on the 'Category' to which the 'Item Purchased' belonged to. This information is directly related to 'Item Purchased' column hence, the table shall be referenced using the 'Item Purchased' column as it would form the Primary Key in the 'ItemCategory' table.

As illustrated earlier, another duplicate copy of the 'shopping_trends_updated' table was created and that copy was renamed as 'ItemCategory'. Then, all other columns except for 'Item Purchased' and 'Category' were removed from this new table as shown:



Next, the table was sorted in ascending order by 'Category' column to see if there are any unnecessary redundancy, but as shown, none are present. So, here, 'Remove Duplicates' cannot be performed as all are unique rows. Hence, the final 'ItemCategory' table is as below:



Once again, the ‘Category’ column was removed from the ‘shopping_trends_updated’ table while retaining the ‘ItemPurchased’ column as it is the Foreign Key.

The screenshot shows the Power BI Desktop interface. The main view displays a table with the following data:

	Gender	Item Purchased	Category	Purchase
1	55 Male	Blouse	Clothing	
2	19 Male	Sweater	Clothing	
3	50 Male	Jeans	Clothing	
4	21 Male	Sandals	Footwear	
5	45 Male	Blouse	Clothing	
6	46 Male	Sneakers	Footwear	
7	63 Male	Shirt	Clothing	
8	27 Male	Shorts	Clothing	
9	26 Male	Coat	Outerwear	
10	57 Male	Handbag	Accessories	
11	53 Male	Shoes	Footwear	
12	30 Male	Shorts	Clothing	
13	61 Male	Coat	Outerwear	
14	65 Male	Dress	Clothing	
15	64 Male	Coat	Outerwear	
16	64 Male	Skirt	Clothing	
17	25 Male	Sunglasses	Accessories	
18	53 Male	Dress	Clothing	

The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list with 'Removed Columns' selected.

c. Creating CustomerInfo Table

The ‘CustomerInfo’ table is used to access the information of each customer based on the ‘Customer ID’. The columns that can be accessed are ‘Age’, ‘Gender’, ‘Location’ and ‘Subscription Status’ and these describes, for each customer, their age, gender, location in the United States and whether they have subscribed or not to the website. This information is directly related to ‘Customer ID’ column hence, the table shall be referenced using the ‘Customer ID’ column as it would form the Primary Key in the ‘CustomerInfo’ table .

The steps for duplicating the ‘shopping_trends_updated’ table were performed and the table was renamed as ‘CustomerInfo’ table. All other columns except for ‘Customer ID’, ‘Age’, ‘Gender’, ‘Location’ and ‘Subscription Status’ were removed to form the following table:

Queries

= Table.SelectRows("Removed Other Columns", each true)

	Customer ID	Age	Gender	Location	Subscription Status
1	1	55	Male	Kentucky	Yes
2	2	19	Male	Maine	Yes
3	3	50	Male	Massachusetts	Yes
4	4	21	Male	Rhode Island	Yes
5	5	45	Male	Oregon	Yes
6	6	46	Male	Wyoming	Yes
7	7	63	Male	Montana	Yes
8	8	27	Male	Louisiana	Yes
9	9	26	Male	West Virginia	Yes
10	10	57	Male	Missouri	Yes
11	11	53	Male	Arkansas	Yes
12	12	30	Male	Hawaii	Yes
13	13	61	Male	Delaware	Yes
14	14	65	Male	New Hampshire	Yes
15	15	64	Male	New York	Yes
16	16	64	Male	Rhode Island	Yes
17	17	25	Male	Alabama	Yes
18	18	53	Male	Mississippi	Yes
19	19	52	Male	Montana	Yes

Since all values were unique, no further steps were performed.

As in previous cases, the same columns were removed from the 'shopping_trends_updated' table. But, the 'Customer ID' column was retained in the 'shopping_trends_updated' table as it is the Foreign Key.

d. Creating PurchaseTrend Table

The 'PurchaseTrend' table is used to access the information of each customer based on the 'Customer ID'. The columns that can be accessed are 'Previous Purchases' and 'Frequency of Purchases' and these describes, for each customer, their number of previous purchases and their frequency of purchasing such as annually, monthly etc. This information is directly related to 'Customer ID' column hence, the table shall be referenced using the 'Customer ID' column as it would form the Foreign Key in the 'PurchaseTrend' table.

The steps for duplicating the 'shopping_trends_updated' table were performed and the table was renamed as 'PurchaseTrend' table. All other columns except for 'Customer ID', 'Previous Purchases' and 'Frequency of Purchases' were removed to form the following table:

Table: SelectColumns(#"Removed Columns",{"Customer ID", "Previous Purchases", "Frequency of Purchases"})

Customer ID	Previous Purchases	Frequency of Purchases
1	1	14 Fortnightly
2	2	2 Fortnightly
3	3	23 Weekly
4	4	49 Weekly
5	5	31 Annually
6	6	14 Weekly
7	7	49 Quarterly
8	8	19 Weekly
9	9	8 Annually
10	10	4 Quarterly
11	11	26 Bi-Weekly
12	12	10 Fortnightly
13	13	37 Fortnightly
14	14	31 Weekly
15	15	34 Weekly
16	16	8 Monthly

This table too featured unique rows.

The same columns were removed from the 'shopping_trends_updated' table. But, the 'Customer ID' column was retained in the 'shopping_trends_updated' table as it is the Foreign Key.

e. Creating ShippingInfo Table

The 'ShippingInfo' table is used to access the information of each customer based on the 'Customer ID'. The column that can be accessed is 'Shipping Type' and this column describes, for each customer, the shipping type used for delivering their purchased product such as express, free shipping etc. This information is directly related to 'Customer ID' column hence, the table shall be referenced using the 'Customer ID' column as it would form the Primary Key in the 'ShippingInfo' table.

The steps for duplicating the 'shopping_trends_updated' table were performed and the table was renamed as 'ShippingInfo' table. All other columns except for 'Customer ID' and 'Shipping Type' were removed to form the following table:

Table: SelectColumns(#"Removed Columns",{"Customer ID", "Shipping Type"})

Customer ID	Shipping Type
1	Express
2	Express
3	Free Shipping
4	Road Day Air
5	Free Shipping
6	Standard
7	Free Shipping
8	Free Shipping
9	Express
10	2-Day Shipping
11	Store Pickup
12	Store Pickup
13	Express

The table was found to have unique rows, hence no further changes were made.

The same columns were removed from the 'shopping_trends_updated' table. But, the 'Customer ID' column was retained in the 'shopping_trends_updated' table as it is a key.

As concluding steps, the 'shopping_trends_updated' table was reordered and the table was renamed as 'Main Table'. The three visuals that follow, show these steps and the final table.

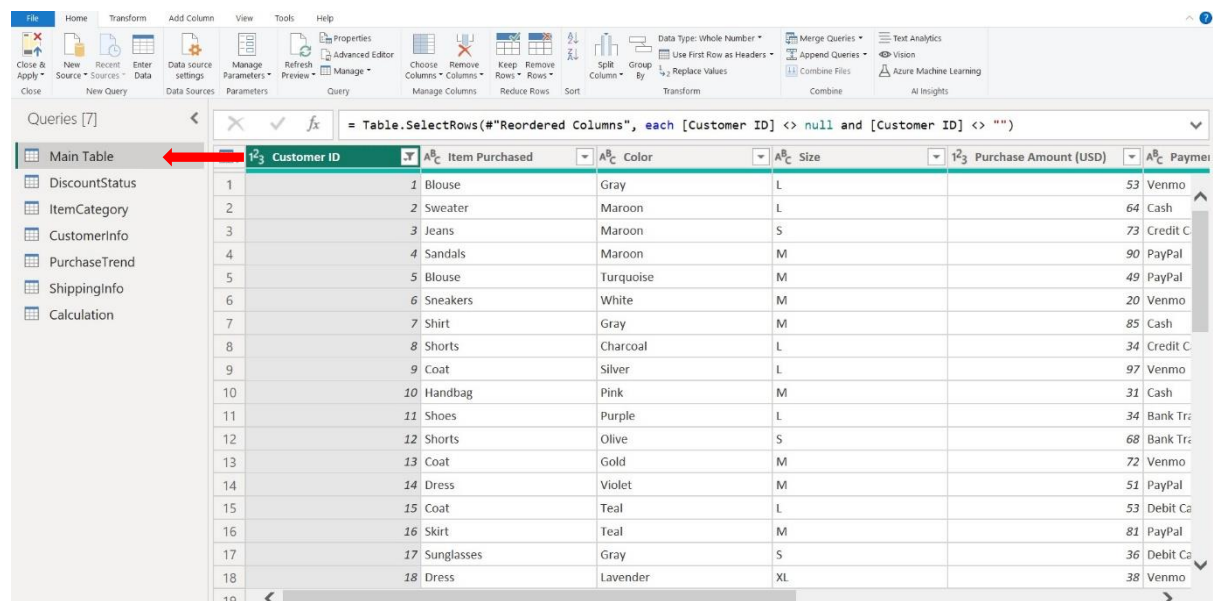
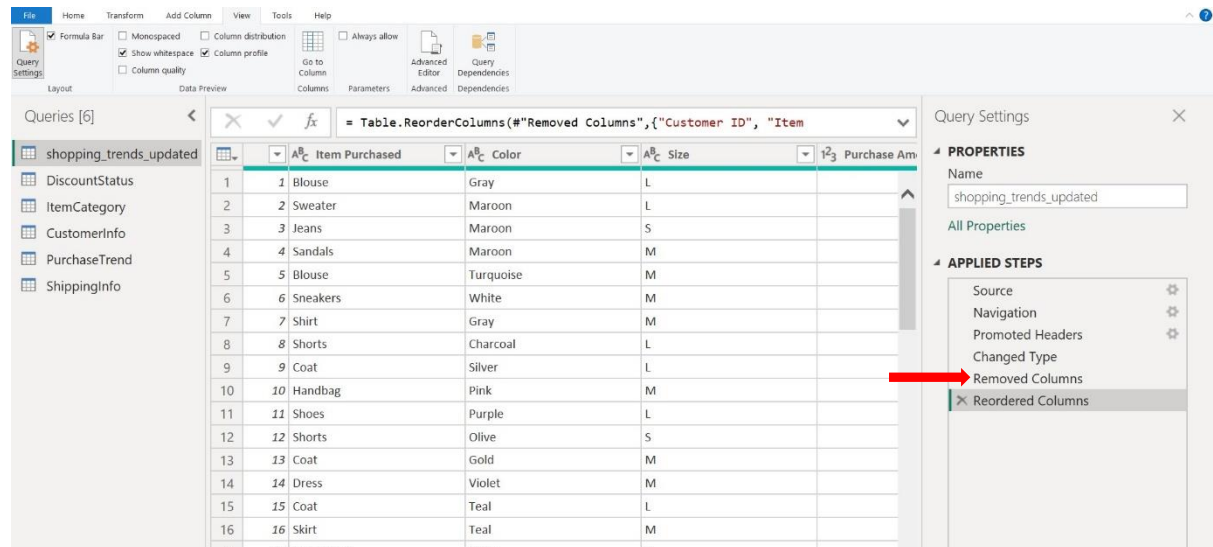


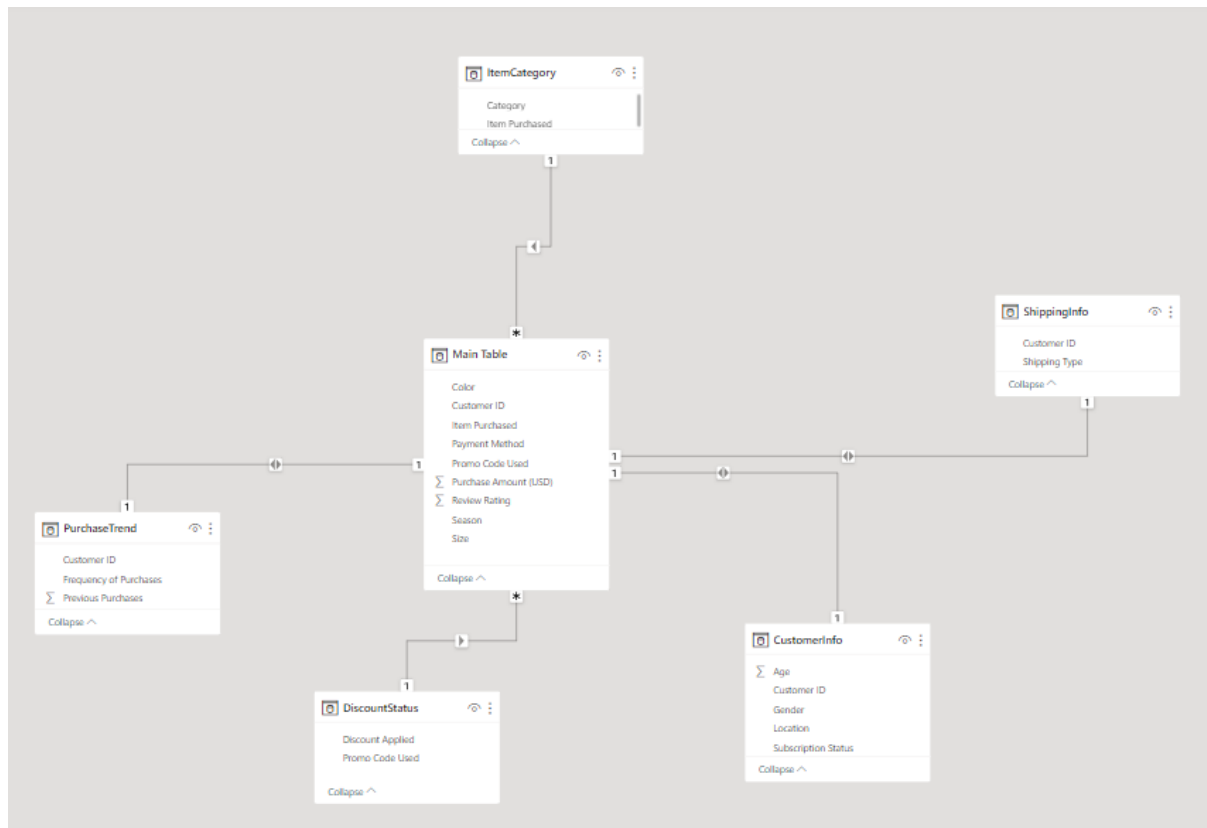
Table.SelectRows("#Reordered Columns", each [Customer ID] <> null and [Customer ID] <> "")								
Item Purchased	Color	Size	Purchase Amount (USD)	Payment Method	Promo Code Used	Season	Review Rating	
1 use	Gray	L	53	Venmo	Yes	Winter	3.1	
2 later	Maroon	L	64	Cash	Yes	Winter	3.1	
3 is	Maroon	S	73	Credit Card	Yes	Spring	3.1	
4 fails	Maroon	M	90	PayPal	Yes	Spring	3.5	
5 use	Turquoise	M	49	PayPal	Yes	Spring	2.7	
6 sakers	White	M	20	Venmo	Yes	Summer	2.9	
7 t	Gray	M	85	Cash	Yes	Fall	3.2	
8 ts	Charcoal	L	34	Credit Card	Yes	Winter	3.2	
9 t	Silver	L	97	Venmo	Yes	Summer	2.6	
10 tlbag	Pink	M	31	Cash	Yes	Spring	4.8	
11 ts	Purple	L	34	Bank Transfer	Yes	Fall	4.1	
12 ts	Olive	S	68	Bank Transfer	Yes	Winter	4.9	
13 t	Gold	M	72	Venmo	Yes	Winter	4.5	
14 is	Violet	M	51	PayPal	Yes	Spring	4.7	
15 t	Teal	L	53	Debit Card	Yes	Winter	4.7	
16 t	Teal	M	81	PayPal	Yes	Winter	2.8	
17 glasses	Gray	S	36	Debit Card	Yes	Spring	4.1	
18 is	Lavender	XL	38	Venmo	Yes	Winter	4.7	
19 later	Black	S	48	Cash	Yes	Summer	4.6	
20 is	Green	M	90	Debit Card	Yes	Summer	3.3	
21 is	Black	M	51	Cash	Yes	Winter	2.8	
22 is	Charcoal	M	62	Debit Card	Yes	Winter	4.1	
23 is	Peach	M	37	Debit Card	Yes	Summer	3.2	
24 is	White	XL	88	Credit Card	Yes	Winter	4.4	
25 et	Green	M	22	Debit Card	Yes	Fall	2.9	
26								

Thus, the normalisation process has been completed.

Creating Relationships and Modelling

As soon as the Fact Table and the Dimensions Tables were finalised, Power BI automatically established relationships between the tables. Upon inspection, the relationships created by default, were the same as the relationships intended to be created.

Thus, the **Star Schema** was established as shown below:



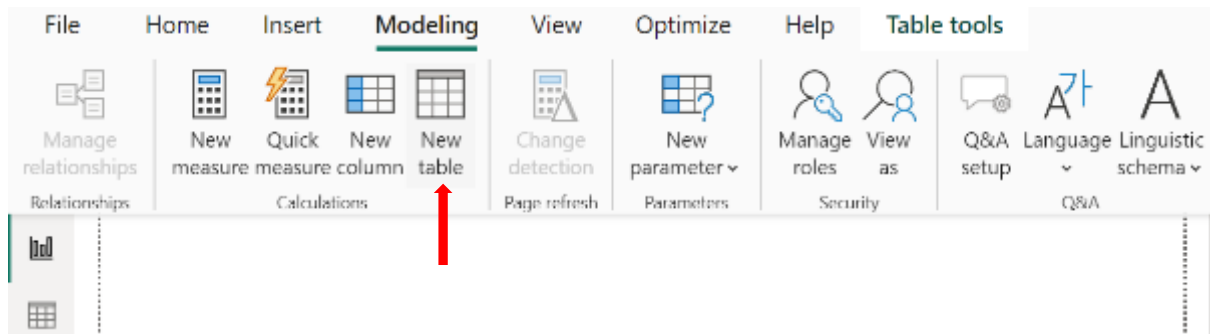
The one-to-one relationships were created between the Facts Table which is titled: 'Main Table' and the Dimensions Tables that shared the key: 'Customer ID'. This is because each row in the 'Main Table' corresponded to each row in all the Dimensions Tables that have one-to-one relationships. Since vice-versa is true, the relationships were rightly bi-directional in cardinality.

Meanwhile, the many-to-one relationships were created between the 'Main Table' (Facts Table) and the Dimensions Tables that had different keys as, multiple rows in the 'Main Table' corresponded single tables in these Dimensions Tables. Thus, a single row in one of the Dimensions Tables were linked to multiple rows in the 'Main Table'. Hence, the relationships were also rightly unidirectional, that is the arrow points from Dimensions Tables to the 'Main Table'.

Creating Measures Table ‘Calculation’

A separate table ‘Calculation’ needs to be created for all the measures to be organised together for easy access.

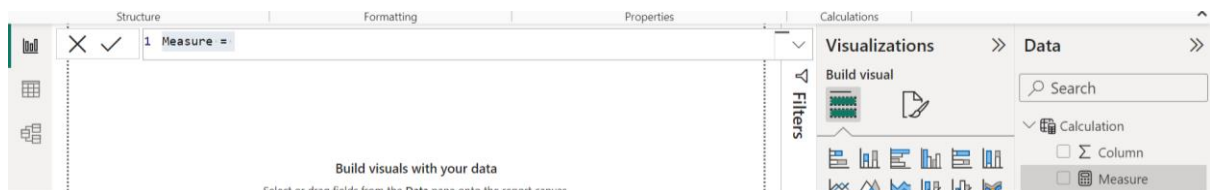
For this, from the ‘Modelling’ tab, the ‘New Table’ option was selected as shown:



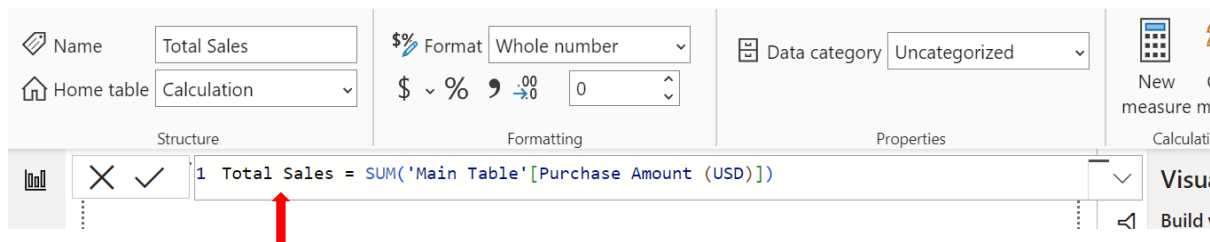
The table, as shown below has an empty column. The table was named: ‘Calculation’:



Next a new measure, for the newly created ‘Calculation’ table was created as shown below:



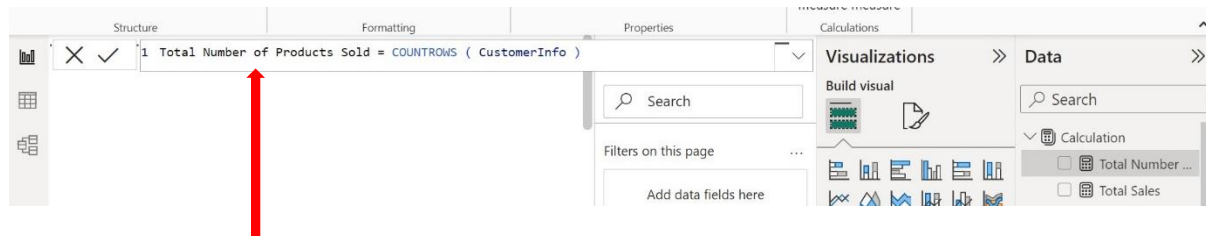
The measure was to calculate ‘Total Sales’ in the ‘Purchase Amount (USD)’ column of ‘Main Table’. The measure formula using DAX is as follows:



The syntax for the DAX function used here is the `SUM('Table_Name'['Column_Name'])`.

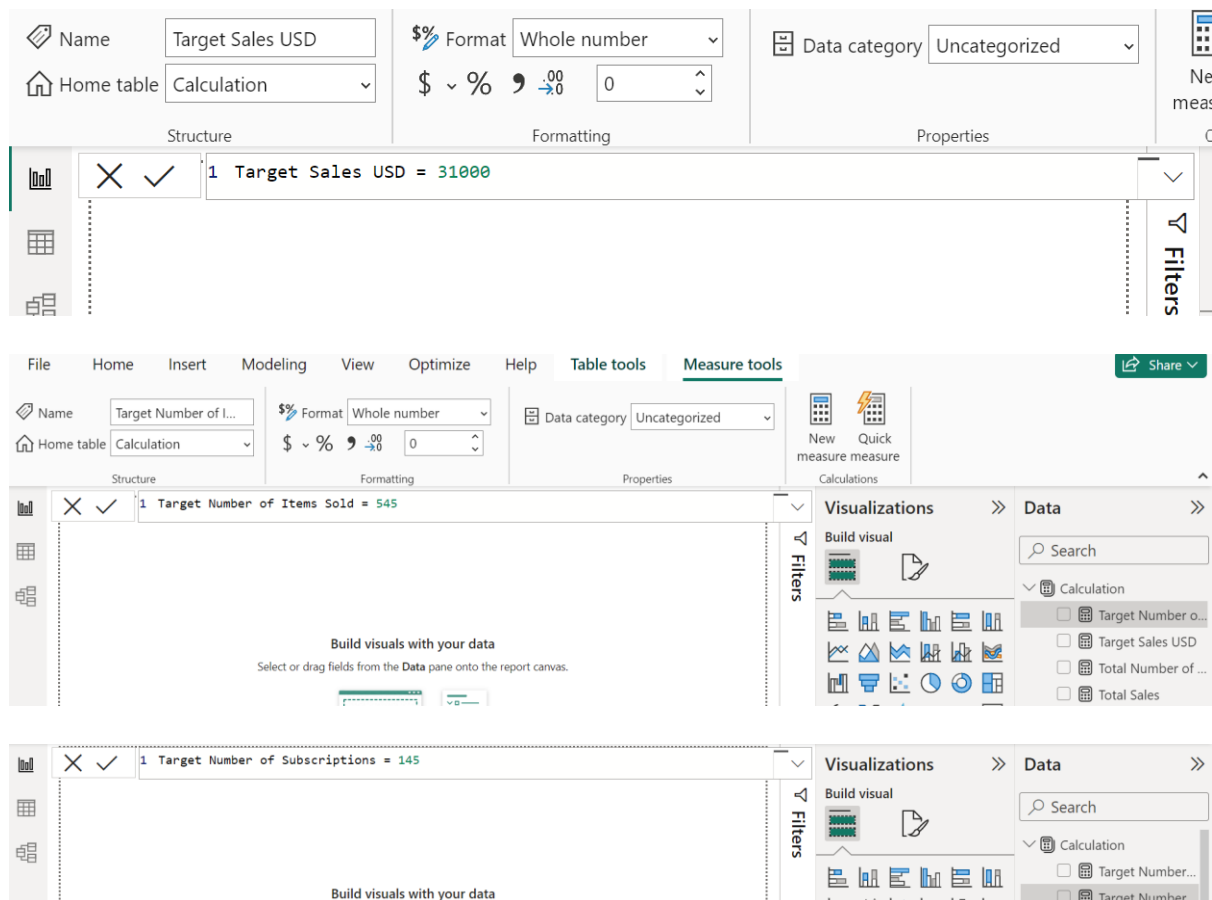
Once this measure was created, the empty column of the 'Calculation' table was deleted. Thus, a table exclusively for measures was created.

Similarly, a second measure was created to find the 'Total Number of Products Sold' in the 'CustomerInfo' table. The measure formula using DAX is as follows:



The syntax for the DAX function used here is the `COUNTROWS('Table_Name')`.

The following measures were created to set 'Target Sales USD', 'Target Number of Items Sold' and 'Target Number of Subscriptions' for defining KPIs later on.



Customer Shopping Trends Analysis

SECTION 2

BUSINESS INTELLIGENCE SOLUTION REPORT

Prepared by:

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For:

CIS4008-N-CF1-2023

BIG DATA AND BUSINESS INTELLIGENCE

January, 2024

Teesside University, United Kingdom

A. Executive Summary

Background

In the present business environment, Retail, e-Commerce and associated sectors are some of the most booming domains worldwide. With several business groups and entrepreneurs venturing into this versatile domain, it is more relevant than ever. Due to the increasing levels of competition in this industry, Data Analytics and Business Intelligence solutions, developed using advanced technology is the key. This report focuses on the general customer shopping trends and preferences observed for a typical retail shopping business that deals with apparel shopping. The report illustrates the key findings and the BI solutions derived from inferences of the Customer Shopping Trends Analysis project. Specific study of Total Sales, Total Unit Sales and Subscription figures have been examined and analysed for most frequent weekly shoppers alongside analysing Item Category popularity, Gender based figures and Size range preferences have also been observed.

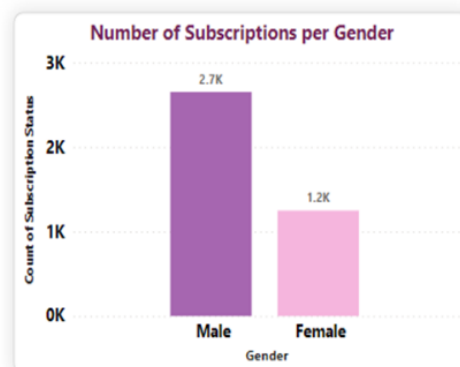
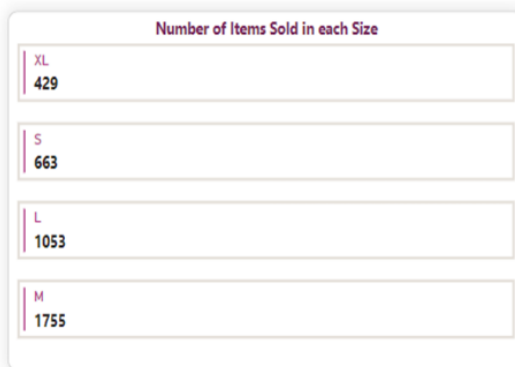
Data Model



Key Findings

- Total Sales observed for customers who has a frequency of shopping weekly is 2.54% higher than the target, indicating that the business has been able to attain its target sales with the weekly customers very well.
- Total Number of Items Sold to customers who has a frequency of shopping is 1.1% lower than the target, indicating that the business can take initiatives to improve unit sales for customers who purchase with a weekly frequency, to meet its targets.

- Total Number of Items Sold to customers who has a frequency of shopping weekly is 8.28% higher than the target, indicating that the business is popular among customers who purchase with a weekly frequency.
- The size range 'M' had the greatest number of items sold, making it the most preferred size range.
- The Male customers have recorded 2.7K Nos. of subscriptions, to be the gender with the greatest number of subscriptions whereas the Female customers have recorded a relatively lower level, which is 1.2K Nos. of subscriptions.
- Total Sales in USD is estimated at 233K, the Total Number of Items Sold is 3900 units and Average Customer Review Rating is 3.75.
- The Sales recorded for subscribed customers is 0.06M USD, whereas, the Sales recorded for customers with no subscription is 0.17M USD.
- In Seasonal Sales analysis, the season 'Fall' saw the greatest amount of Total Sales.





Recommendations

- Total Number of Items sold to customers who has a frequency of shopping Weekly needs to be improved to meet the set goal.
- The lower levels of Female subscriptions may be investigated and boosted while utilizing the current levels of Male subscriber base, which can help with improvement in sales performance.
- The Size range 'M', being the most preferred size, would require restocking.
- The 'Fall' season is the right time to release new products as this is the season where the majority sales are likely to happen.

B. Introduction

The retail industry is one of the most competitive domains, that requires constant monitoring alongside being continuously updated on the latest trends in the industry. A very important defining factor in the retail domain, for any business, is its customer base and clientele.

The ‘Customer Shopping Trends Analysis’ project aims to study and visualise a typical apparel business using the powerful BI tool ‘PowerBI’. The dataset chosen for the same features the following data columns: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Colour, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, Payment Method, Frequency of Purchases.

The **dataset** has been obtained from the following source:

<https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset/data>

The **data model** created for gaining insight into the data are as follows:

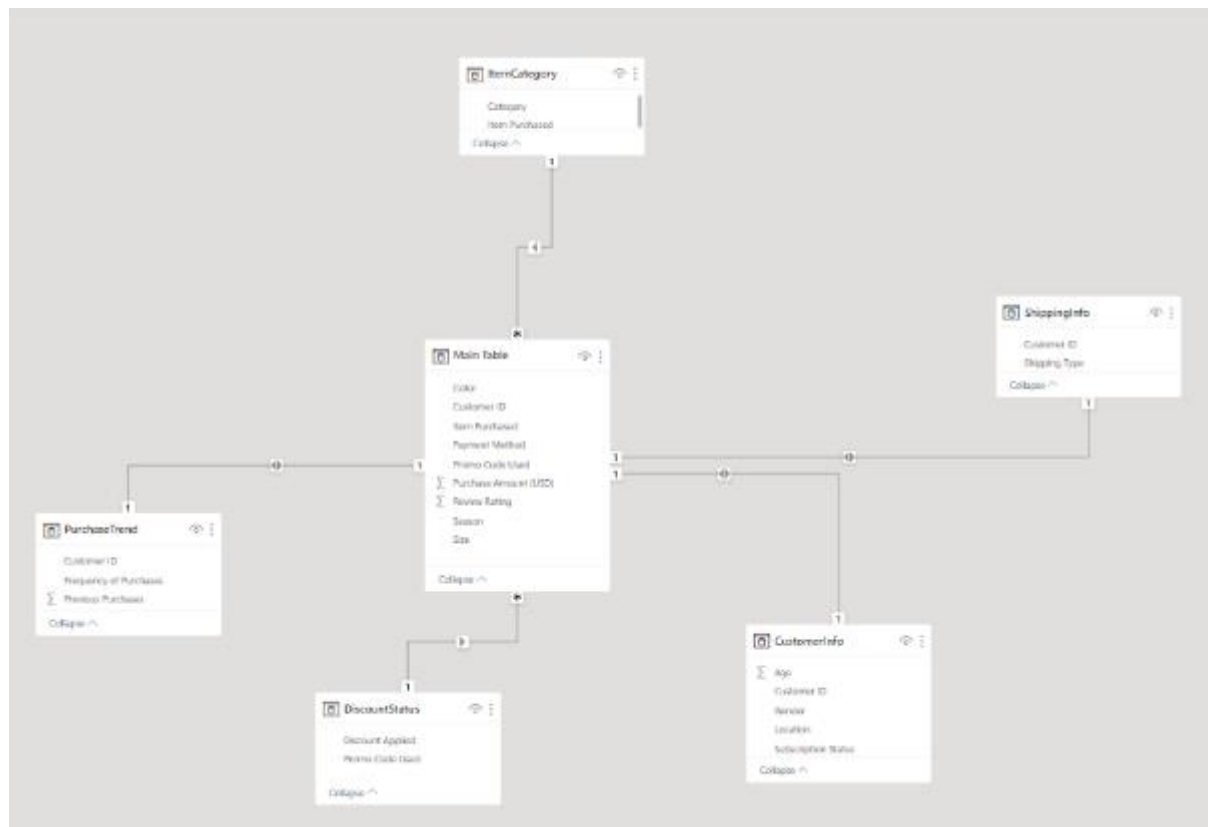


Figure 2: Data Model

The model features a fact table: Main Table and five dimension tables: DiscountStatus, ItemCategory, CustomerInfo, PurchaseTrend and ShippingInfo. These tables provide detailed information of each unique customer, sold item details etc. as discussed in section 1.

The report addresses the following important Business Intelligence Questions:

- a. What is the Total Sales Observed for customers who has a frequency of shopping Weekly?
- b. What is the Total Number of Items sold to customers who has a frequency of shopping Weekly?
- c. What is the Number of Subscriptions observed for customers who has a frequency of shopping Weekly?
- d. Which is the most sold Item to customers who has a frequency of shopping Annually?
- e. How many items were sold in each Size range?
- f. Which Gender type has recorded the greatest number of subscriptions?
- g. What is the Total Sales observed, Total Number of Items sold and Average Customer Review Rating?
- h. What is the Total Sales recorded for customers with and without Subscription?
- i. Which Season saw the greatest amount of Total Sales?

C. Key Findings based on Evaluation and Analysis

The primary focus of this analysis is on the performance of the business alongside understanding customer shopping behaviour, preferences and sales trends considering variables such as season, size range of items etc.

The dashboard, created to visualise and interact this data is as follows:

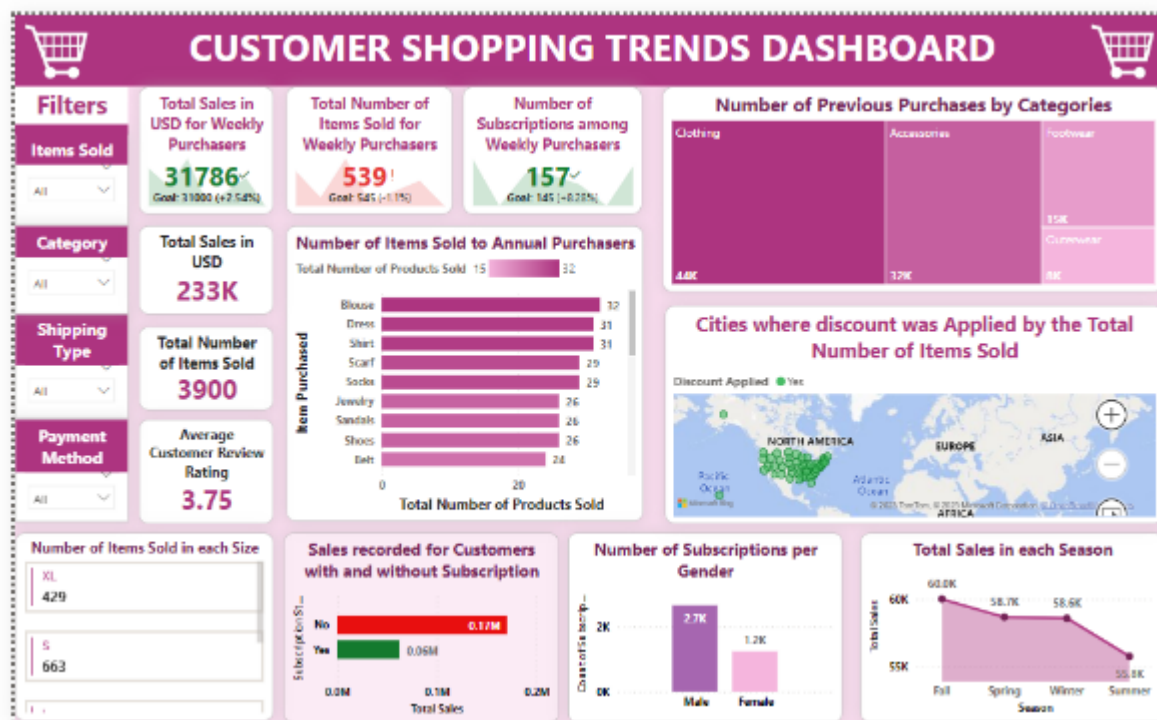


Figure 3: Customer Shopping Trends Dashboard

The dashboard consists of interactive charts that can be filtered using the 'Filter' pane, based on the fields: 'Item Sold', 'Category', 'Shipping Type' and 'Payment Method'. The dashboard also features KPI visuals that can be used to track the progress of the specific quantities. The KPIs are unaffected by the filtering.

The KPIs selected are: 'Total Sales in USD for Weekly Purchasers', 'Total Number of Items Sold for Weekly Purchasers' and 'Number of Subscriptions among Weekly Purchasers'. These KPIs were chosen and specifically designed for customers with a frequency of purchasing weekly as this clientele can be considered as the most frequent and potentially, loyal group of customers. The aim is to maintain this clientele for the longest of time, as they form the most valuable and crucial customer base.

A) What is the Total Sales Observed for customers who has a frequency of shopping Weekly?



This is a KPI visual, created to analyse the total sales by customers who has a frequency of purchasing weekly. Here, as discussed in section 1, the 'Total Sales' measure has been used to calculate the sales value for customers who shop with a weekly frequency. An estimated goal value of 31000 USD was pre-set by using the 'Target Sales USD' measure, as discussed in section 1.

As per the KPI visual, the measured total sales recorded for customers who purchase with a weekly frequency is 31786 USD, which is 2.54% higher than the target of 31000 USD.

Thus, it can be concluded that the Total Sales observed for customers who has a frequency of shopping weekly is 31786 USD and it is 2.54% higher than the target, indicating that the business has been able to attain its target sales with the weekly customers very well.

B) What is the Total Number of Items sold to customers who has a frequency of shopping Weekly?

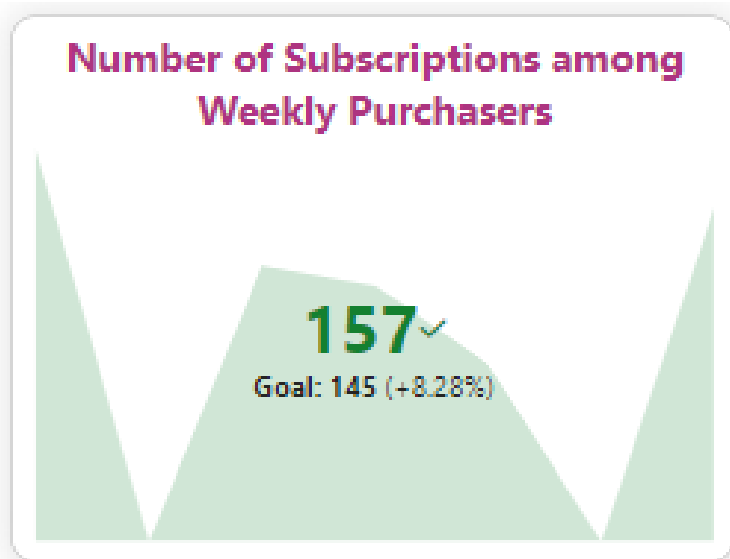


This KPI visual, has been created to analyse the total number of products sold to customers who has a frequency of purchasing weekly. Here, as discussed in section 1, the ‘Total Number of Products Sold’ measure has been used to calculate the sales value for customers who shop with a weekly frequency. An estimated goal value of 545 units was pre-set by using the ‘Target Number of Items Sold’ measure, as discussed in section 1.

As per the KPI visual, the measured total number of items sold recorded for customers who purchase with a weekly frequency is 539 units, which is 1.1% lower than the target of 545 units.

Thus, it can be concluded that the Total Number of Items Sold to customers who has a frequency of shopping weekly is 539 units and it is 1.1% lower than the target, indicating that the business can take initiatives to improve unit sales for customers who purchase with a weekly frequency, to meet its targets.

C) What is the Number of Subscriptions observed for customers who has a frequency of shopping Weekly?

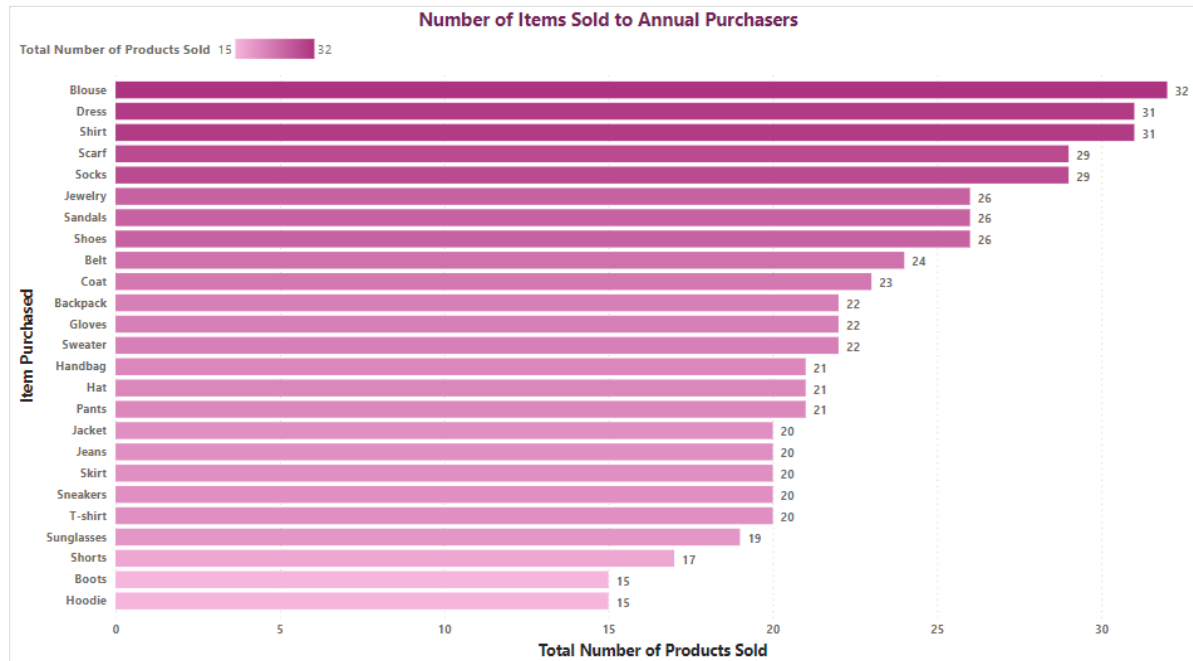


This KPI visual, has been created to analyse the total number of subscriptions among customers who has a frequency of purchasing weekly. Here, as discussed in section 1, the ‘Total Number of Products Sold’ measure has been used to calculate the sales value for customers who shop with a weekly frequency. An estimated goal value of 145 Nos. was pre-set by using the ‘Target Number of Subscriptions’ measure, as discussed in section 1.

As per the KPI visual, the measured total number of items sold recorded for customers who purchase with a weekly frequency is 157 Nos., which is 8.28% higher than the target of 145 Nos.

Thus, it can be concluded that the Total Number of Items Sold to customers who has a frequency of shopping weekly is 157 Nos. and it is 8.28% higher than the target, indicating that the business is popular among customers who purchase with a weekly frequency.

D) Which is the most sold Item to customers who has a frequency of shopping Annually?



This visual, created using a Horizontal Bar Chart, displays the name of Item Purchased on the Y axis and the Number of Products Sold for that particular Item type. The 'Total Number of Products Sold' measure has been used for this visual. This visual has been affected by the 'Purchase Frequency' filter applied and thus shows the data for only customers who have a frequency of shopping annually.

According to this chart, it can be concluded that the Item 'Blouse' has been the most sold item to customers who have a frequency of shopping Annually. The number of products sold for the Item 'Blouse' is 32 units.

This information is useful to determine which item needs to be marketed more to customers who only shop once a year. This can also help plan inventory and restocking for the business.

E) How many Items were sold in each Size range?



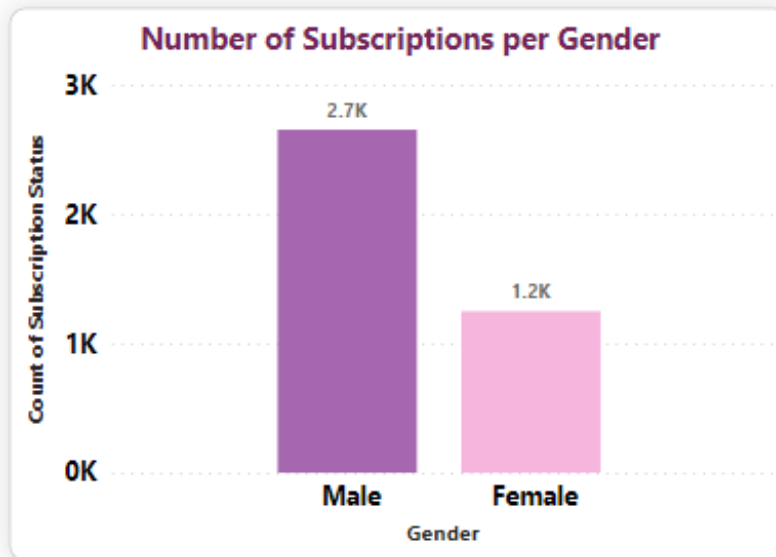
This chart is based on a unique visual known as multi-row card, that can be used to analyse data with limited sub categories such as size range.

The chart provides information on the number of sold items belonging to each size range. The size ranges offered by this business are: S, M, L and XL. The chart has been ordered in a downward ascending order, so that the size range with the least number of items sold appears first and the size range with the greatest number of items sold appears last. The measure 'Total Number of Products Sold' has been used in this visual.

This chart can be used to assess general size preferences of customers and to make decisions on restocking of items in these size ranges. The visual can also be used to analyse the number of products sold in each size for different filter categories as well.

Thus, to conclude, the overall number of items sold in each Size range has been visualised accurately. It is also inferred that the size range XL had the least number of items sold, whereas the size range M is had the greatest number of items sold, making it the most preferred size range and XL, the least preferred, among customers.

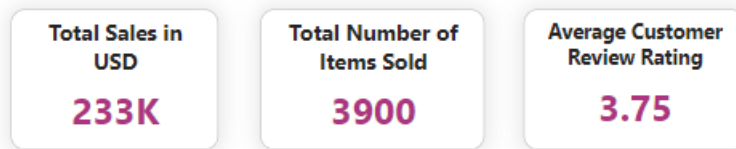
F) Which Gender type has recorded the greatest number of Subscriptions?



This vertical column chart is the best choice to visualise the difference in number of subscriptions between the two genders.

The Male customers have recorded 2.7K Nos. of subscriptions, whereas the Female customers have recorded a relatively lower level, which is 1.2K Nos. of subscriptions. To conclude, the business may take initiatives to take maximum advantage of its Male subscriptions to improve its business performance while investigating why there could be a lesser number of Female subscribers. It can, in fact, take actions to improve its Female customer base.

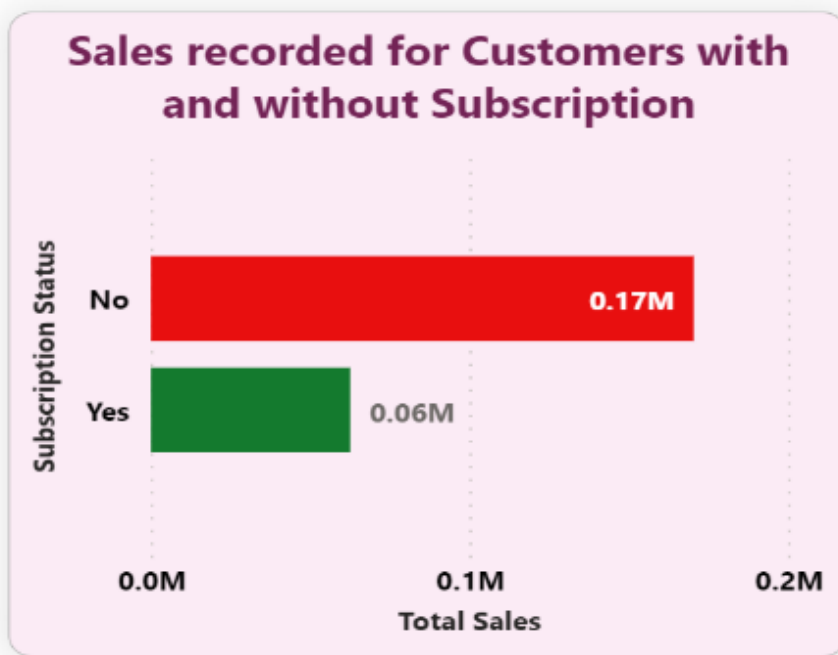
G) What is the Total Sales observed, Total Number of Items sold and Average Customer Review Rating?



As per the visuals generated, the Total Sales in USD is estimated at 233K, the Total Number of Items Sold is 3900 units and Average Customer Review Rating is 3.75. These visuals have been generated using ‘Card’ type charts in PowerBI. The values are estimated for the entire list of customers. But these visuals can also display the concerned values when the aforementioned filters on the page are applied.

The conclusion is that, the Total Sales in USD, Total Number of Items Sold and Average Customer Review Rating have been visualised for the entire list of customers.

H) What is the Total Sales recorded for Customers with and without Subscription?



The horizontal bar graph shown above visualises the Total Sales (USD) for each type of Subscription Status of customers. The two responses are: 'Yes', indicating customers who have subscription and 'No', indicating customers with no subscription.

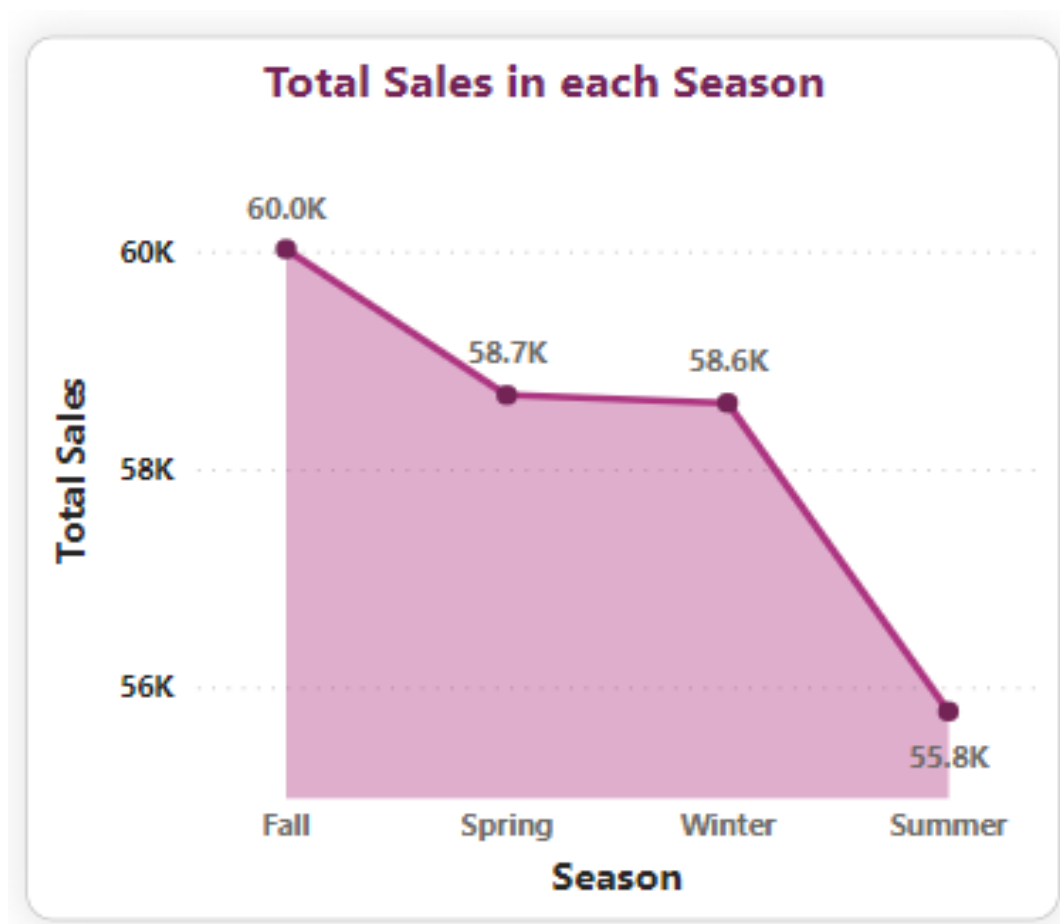
This analysis has actually presented a very interesting result. The customers who have not subscribed to the business tend to have purchased items worth 0.17M USD worth in sales meanwhile, customers who have a subscription, surprisingly account for a relatively lower value of sales which is 0.06M USD.

This may be interpreted in several ways. Since, subscription is a measure of popularity among the customers of the organisation, these results may indicate that the business is now attracting more purchases outside the subscription list. Another aspect would be to investigate on why the sales from the subscribers are lower than expected. The answer to this question

will influence the key decisions to be made regarding the business and its initiatives to boost sales and unit sales.

Thus, to conclude, the Sales recorded for subscribed customers is 0.06M USD, whereas, the Sales recorded for customers with no subscription is 0.17M USD. These, results can be further studied to gather several important insights.

I) Which Season saw the greatest amount of Total Sales?



The Total Sales vs Season visual has been created using the Area Chart visual on PowerBI. The inferences from the chart are that the season 'Fall' has recorded the greatest Total Sales of above 60.0K USD, meanwhile, the season 'Summer' saw the least amount of Total Sales at around 55.8K USD.

This information would prove beneficial for deciding new product releases, announcement of offers and scheduling of important business operations. The filters on the dashboard help answer the same questions when the visual is filtered based on the various filter categories.

To conclude, the season 'Fall' saw the greatest amount of Total Sales.

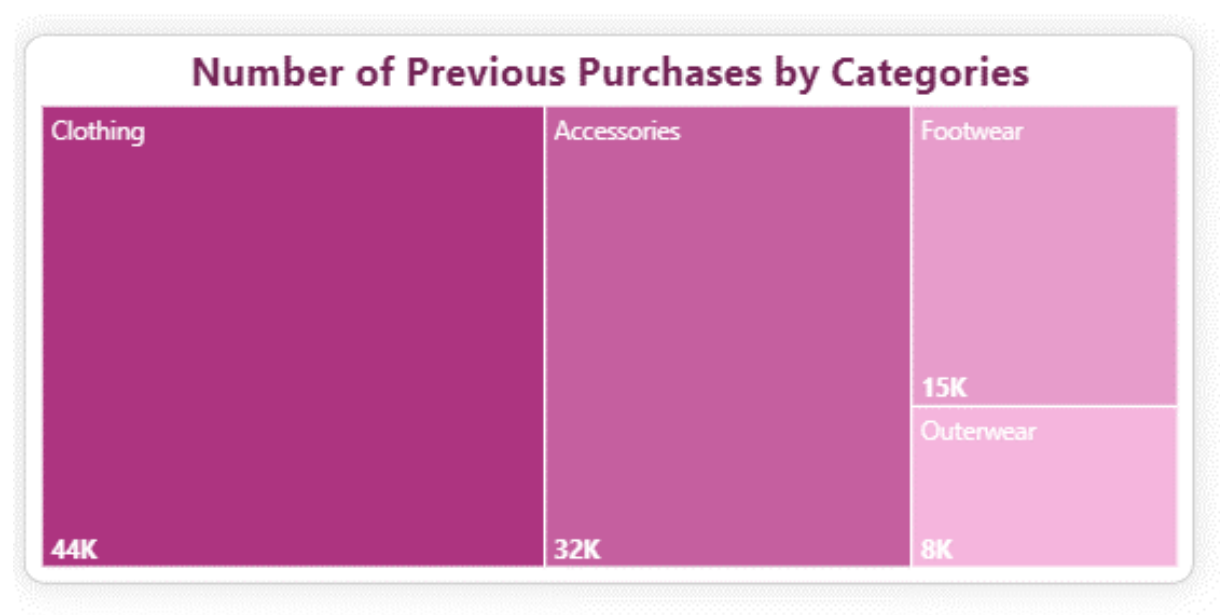
Additional Charts

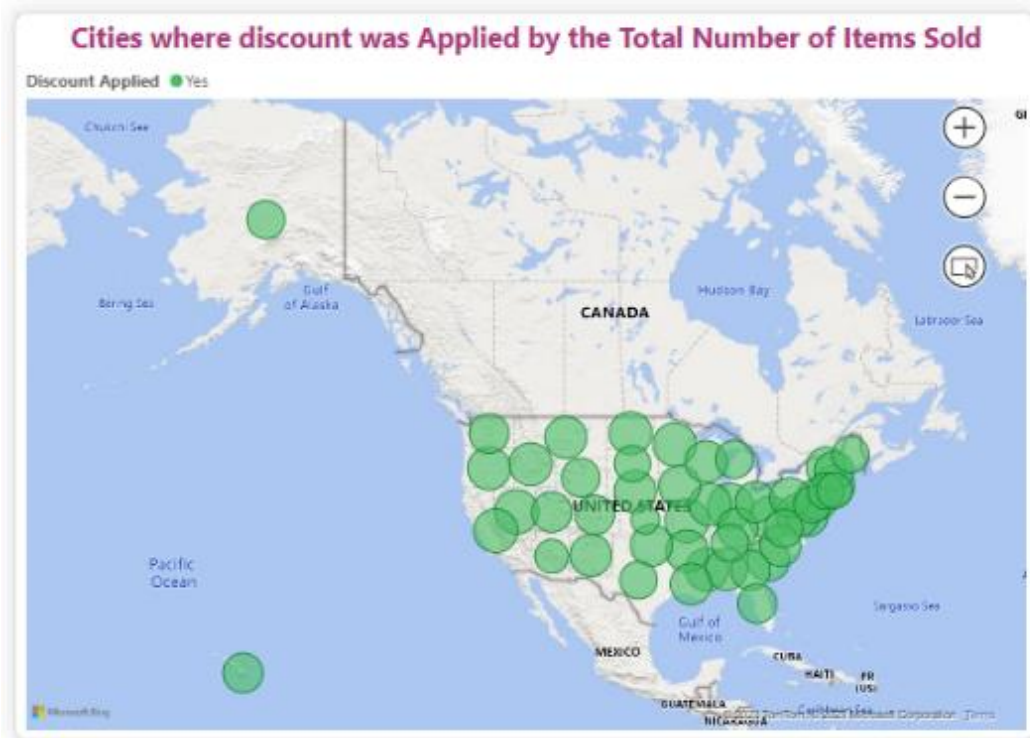
Given are the two additional charts to be used for understanding further details.

This first chart, which is as shown below, illustrates the purchase history of each Category. The visual shows here that, the category 'Clothing' was the most purchased previous category with more than 44K number of previous purchases. Whereas, the category, 'Outerwear' is the least purchased category with number of previous purchases at 8K.

The 'Treemap' visual from PowerBI has been used to create this chart hence it is very self-explanative in nature.

The purchase history can help understand the popularity of the categories available. If the visuals are filtered in the dashboard, the most preferred category for each filtering can be obtained to make informed business decisions.





The second chart, as given above, is a location map-based chart. The chart visualises the different cities of the customers on the physical map of United States of America. The bubbles, point at locations of the customers in USA. It must be noted that only customers who had purchased the item after applying a discount, are represented in this visual. The size of the bubbles represents the number of such items sold to that particular location after the application of discount. The PowerBI dashboard enables viewing the details on this map by simply hovering the pointer on each bubble. This visual can be filtered using the dashboard filters so as to view this visual for different field categories.

Information regarding the location of customers, who applied discount, would be beneficial to plan discount rates and offers in those regions in the future.

D. Conclusions

The Customer Shopping Analysis project was commenced with the aim to study and identify the characteristics of a retail business in terms of the customer preferences, shopping behaviours and sales trends also identify the business intelligence questions that need to be addressed for this particular case. For this purpose, the dataset of a typical retail business was selected. The dataset featured details of each unique customer and the specific attributes defined for each customer were Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Colour, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, Payment Method and Frequency of Purchases. To address the defined business intelligence questions based on the dataset, the versatile BI tool, PowerBI has been employed. The data was normalised and modelled to form a fact table and five dimensions table. Furthermore, calculations were carried out by defining specific calculation measures.

Proceeding further, Total Sales (USD), Total Number of Items Sold and Number of Subscriptions observed for customers with a purchasing frequency of Weekly were defined as the Key Performance Indicators (KPIs). To gain deeper insights, several visualisations were designed as a part of an interactive dashboard. The following were the key observations and insights based on the BI questions addressed:

- Total Sales observed for customers who has a frequency of shopping weekly is 31,786 USD and it is 2.54% higher than the target, indicating that the business has been able to attain its target sales with the weekly customers very well.
- Total Number of Items Sold to customers who has a frequency of shopping weekly is 539 units and it is 1.1% lower than the target, indicating that the business can take initiatives to improve unit sales for customers who purchase with a weekly frequency, to meet its targets.
- Total Number of Items Sold to customers who has a frequency of shopping weekly is 157 Nos. and it is 8.28% higher than the target, indicating that the business is popular among customers who purchase with a weekly frequency.
- The Item 'Blouse' has been the most sold item to customers who have a frequency of shopping Annually. The number of products sold for the Item 'Blouse' is 32 units.

- The overall number of items sold in each Size range has been visualised accurately. It is also inferred that the size range XL had the least number of items sold, whereas the size range M is had the greatest number of items sold, making it the most preferred size range and XL, the least preferred, among customers.
- The Male customers have recorded 2.7K Nos. of subscriptions, whereas the Female customers have recorded a relatively lower level, which is 1.2K Nos. of subscriptions, making out the Male category to be the gender with the most number of subscriptions.
- Total Sales in USD is estimated at 233K, the Total Number of Items Sold is 3900 units and Average Customer Review Rating is 3.75.
- The Sales recorded for subscribed customers is 0.06M USD, whereas, the Sales recorded for customers with no subscription is 0.17M USD.
- In Seasonal Sales analysis, the season 'Fall' saw the greatest amount of Total Sales.
- Additional information on the Purchase History per Category of Item sold and the locations of customers who applied a discount for their purchase can also be obtained and visualised with the aid of the dashboard.

E. Recommendations

Based on the key insights and inferences obtained from the dashboard, that has been designed to address the earlier mentioned BI questions, the following recommendations can be suggested:

- Total Number of Items sold to customers who has a frequency of shopping Weekly needs to be improved to meet the set goal. This maybe carried out through efficient marketing by targeting the right audience and customer base.
- The lower levels of Female subscriptions may be investigated and boosted while utilizing the current levels of Male subscriber base, which can help with improvement in sales performance.
- The Size range 'M', being the most preferred size, would require restocking. The number of items sold in each Size range can in fact help with inventory planning and procurement processes.
- The 'Fall' season is the right time to release new products as this is the season where the majority sales are likely to happen. Furthermore, detailed market analysis can help decide the kind of products that needs to be introduced.
- The surprising results of the visualisation of the Total Sales recorded for customers who have subscribed to the business and the customers who do not have a subscription with the business, needs to be examined on a deeper level, as the sales levels being relatively lower for customers with subscription could indicate a change in preferences of the customers who subscribed. This can be analysed further to make alterations in the business policies and operations if necessary.

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